

Solutions to Exercise 2A

1 **A**When $n = 1$,

$$\begin{aligned}\text{RHS} &= 1^2 \\ &= 1 \\ &= \text{LHS}\end{aligned}$$

so the statement is true for $n = 1$.**B**Suppose that $k \geq 1$ is a positive integer for which the statement is true.

$$\text{That is, suppose } 1 + 3 + 5 + \cdots + (2k - 1) = k^2 \quad (**)$$

We prove the statement for $n = k + 1$.

$$\text{That is, we prove } 1 + 3 + 5 + \cdots + (2k - 1) + (2(k + 1) - 1) = (k + 1)^2$$

$$\begin{aligned}\text{LHS} &= 1 + 3 + 5 + \cdots + (2k - 1) + (2(k + 1) - 1) \\ &= 1 + 3 + 5 + \cdots + (2k - 1) + (2k + 2 - 1) \\ &= 1 + 3 + 5 + \cdots + (2k - 1) + (2k + 1) \\ &= k^2 + (2k + 1) && \text{by the induction hypothesis (**)} \\ &= k^2 + 2k + 1 \\ &= (k + 1)^2 \\ &= \text{RHS}\end{aligned}$$

CIt follows from parts **A** and **B** by mathematical induction, that:

$$1 + 3 + 5 + \cdots + (2n - 1) = n^2 \text{ for all integers } n \geq 1.$$

2a **A**When $n = 1$,

$$\begin{aligned}\text{RHS} &= \frac{1}{2}(1)(1 + 1) \\ &= 1 \\ &= \text{LHS}\end{aligned}$$

so the statement is true for $n = 1$.**B**Suppose that $k \geq 1$ is a positive integer for which the statement is true.

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That is, suppose $1 + 2 + 3 + \dots + k = \frac{1}{2}k(k+1)$ (**)

We prove the statement for $n = k + 1$.

That is, we prove $1 + 2 + 3 + \dots + k + (k + 1) = \frac{1}{2}(k + 1)((k + 1) + 1)$

$$\begin{aligned}
 \text{LHS} &= 1 + 2 + 3 + \dots + k + (k + 1) \\
 &= \frac{1}{2}k(k + 1) + (k + 1) && \text{by the induction hypothesis (**)} \\
 &= (k + 1)\left(\frac{1}{2}k + 1\right) \\
 &= \frac{1}{2}(k + 1)(k + 2) \\
 &= \frac{1}{2}(k + 1)((k + 1) + 1) \\
 &= \text{RHS}
 \end{aligned}$$

C

It follows from parts **A** and **B** by mathematical induction, that:

$$1 + 2 + 3 + \dots + n = \frac{1}{2}n(n + 1) \text{ for all integers } n \geq 1.$$

2b **A**

When $n = 1$,

$$\begin{aligned}
 \text{RHS} &= 2^1 - 1 \\
 &= 1 \\
 &= \text{LHS}
 \end{aligned}$$

so the statement is true for $n = 1$.

B

Suppose that $k \geq 1$ is a positive integer for which the statement is true.

That is, suppose $1 + 2 + 2^2 + \dots + 2^{k-1} = 2^k - 1$ (**)

We prove the statement for $n = k + 1$.

That is, we prove $1 + 2 + 2^2 + \dots + 2^{k-1} + 2^{(k+1)-1} = 2^{k+1} - 1$

$$\begin{aligned}
 \text{LHS} &= 1 + 2 + 2^2 + \dots + 2^{k-1} + 2^{(k+1)-1} \\
 &= 2^k - 1 + 2^{(k+1)-1} && \text{by the induction hypothesis (**)} \\
 &= 2^k + 2^k - 1 \\
 &= 2 \times 2^k - 1 \\
 &= 2^{k+1} - 1 \\
 &= \text{RHS}
 \end{aligned}$$

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CIt follows from parts **A** and **B** by mathematical induction, that:

$$1 + 2 + 2^2 + \cdots + 2^{k-1} + 2^{n-1} = 2^n - 1 \text{ for all integers } n \geq 1.$$

2c **A**When $n = 1$,

$$\begin{aligned} \text{RHS} &= \frac{1}{4}(5^1 - 1) \\ &= \frac{1}{4}(5 - 1) \\ &= \frac{4}{4} \\ &= 1 \\ &= \text{LHS} \end{aligned}$$

so the statement is true for $n = 1$.**B**Suppose that $k \geq 1$ is a positive integer for which the statement is true.

$$\text{That is, suppose } 1 + 5 + 5^2 + \cdots + 5^{k-1} = \frac{1}{4}(5^k - 1) \quad (**)$$

We prove the statement for $n = k + 1$.

$$\text{That is, we prove } 1 + 5 + 5^2 + \cdots + 5^{k-1} + 5^{(k+1)-1} = \frac{1}{4}(5^{k+1} - 1)$$

$$\begin{aligned} \text{LHS} &= 1 + 5 + 5^2 + \cdots + 5^{k-1} + 5^{(k+1)-1} \\ &= \frac{1}{4}(5^k - 1) + 5^{(k+1)-1} && \text{by the induction hypothesis (**)} \\ &= \frac{1}{4}(5^k - 1 + 4 \times 5^{(k+1)-1}) \\ &= \frac{1}{4}(5^k - 1 + 4 \times 5^k) \\ &= \frac{1}{4}(5 \times 5^k - 1) \\ &= \frac{1}{4}(5^{k+1} - 1) \\ &= \text{RHS} \end{aligned}$$

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C

It follows from parts **A** and **B** by mathematical induction, that:

$$1 + 5 + 5^2 + \cdots + 5^{n-1} = \frac{1}{4}(5^n - 1) \text{ for all integers } n \geq 1.$$

2d A

When $n = 1$,

$$\begin{aligned} \text{RHS} &= \frac{1}{3}(1)(1+1)(1+2) \\ &= \frac{1}{3} \times 2 \times 3 \\ &= 1 \times 2 \\ &= \text{LHS} \end{aligned}$$

so the statement is true for $n = 1$.

B

Suppose that $k \geq 1$ is a positive integer for which the statement is true.

$$\text{That is, suppose } 1 \times 2 + 2 \times 3 + 3 \times 4 + \cdots + k(k+1) = \frac{1}{3}k(k+1)(k+2) \quad (**)$$

We prove the statement for $n = k + 1$.

$$\begin{aligned} \text{That is, we prove } 1 \times 2 + 2 \times 3 + 3 \times 4 + \cdots + k(k+1) + (k+1)((k+1)+1) \\ = \frac{1}{3}(k+1)((k+1)+1)((k+1)+2) \end{aligned}$$

$$\begin{aligned} \text{LHS} &= 1 \times 2 + 2 \times 3 + 3 \times 4 + \cdots + k(k+1) + (k+1)((k+1)+1) \\ &= \frac{1}{3}k(k+1)(k+2) + (k+1)((k+1)+1) \end{aligned}$$

by the induction hypothesis (**)

$$\begin{aligned} &= \frac{1}{3}k(k+1)(k+2) + (k+1)(k+2) \\ &= (k+1)(k+2) \left(\frac{1}{3}k + 1 \right) \\ &= \frac{1}{3}(k+1)(k+2)(k+3) \\ &= \frac{1}{3}(k+1)((k+1)+1)((k+1)+2) \\ &= \text{RHS} \end{aligned}$$

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CIt follows from parts **A** and **B** by mathematical induction, that:

$$1 \times 2 + 2 \times 3 + 3 \times 4 + \cdots + n(n+1) = \frac{1}{3}n(n+1)(n+2) \text{ for all integers } n \geq 1.$$

2e **A**When $n = 1$,

$$\begin{aligned} \text{RHS} &= \frac{1}{6}(1)(1+1)(2+7) \\ &= \frac{1}{6} \times 1 \times 2 \times 9 \\ &= \frac{18}{6} \\ &= 3 \\ &= 1 \times 3 \\ &= \text{LHS} \end{aligned}$$

so the statement is true for $n = 1$.**B**Suppose that $k \geq 1$ is a positive integer for which the statement is true.

That is, suppose:

$$1 \times 3 + 2 \times 4 + 3 \times 5 + \cdots + k(k+2) = \frac{1}{6}k(k+1)(2k+7) \quad (**)$$

We prove the statement for $n = k+1$.

$$\begin{aligned} \text{That is, we prove } 1 \times 3 + 2 \times 4 + 3 \times 5 + \cdots + k(k+2) + (k+1)((k+1)+2) \\ = \frac{1}{6}(k+1)((k+1)+1)(2(k+1)+7) \end{aligned}$$

$$\begin{aligned} \text{LHS} &= 1 \times 2 + 2 \times 3 + 3 \times 4 + \cdots + k(k+2) + (k+1)((k+1)+2) \\ &= \frac{1}{6}k(k+1)(2k+7) + (k+1)((k+1)+2) \end{aligned}$$

by the induction hypothesis (**)

$$\begin{aligned} &= \frac{1}{6}k(k+1)(2k+7) + (k+1)(k+3) \\ &= (k+1) \left(\frac{1}{6}k(2k+7) + k+3 \right) \\ &= \frac{1}{6}(k+1)(k(2k+7) + 6k+18) \\ &= \frac{1}{6}(k+1)(2k^2 + 7k + 6k + 18) \end{aligned}$$

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$$\begin{aligned}
 &= \frac{1}{6}(k+1)(2k^2 + 13k + 18) \\
 &= \frac{1}{6}(k+1)(k+2)(2k+9) \\
 &= \frac{1}{6}(k+1)((k+1)+1)(2(k+1)+7) \\
 &= \text{RHS}
 \end{aligned}$$

C

It follows from parts **A** and **B** by mathematical induction, that:

$$1 \times 3 + 2 \times 4 + 3 \times 5 + \cdots + n(n+2) = \frac{1}{6}n(n+1)(2n+7) \text{ for all integers } n \geq 1.$$

2f **A**

When $n = 1$,

$$\begin{aligned}
 \text{RHS} &= \frac{1}{6}(1)(1+1)(2(1)+1) \\
 &= \frac{1}{6} \times 1 \times 2 \times 3 \\
 &= 1 \\
 &= \text{LHS}
 \end{aligned}$$

so the statement is true for $n = 1$.

B

Suppose that $k \geq 1$ is a positive integer for which the statement is true.

$$\text{That is, suppose } 1^2 + 2^2 + 3^2 + \cdots + k^2 = \frac{1}{6}k(k+1)(2k+1) \quad (**)$$

We prove the statement for $n = k + 1$.

$$\begin{aligned}
 \text{That is, we prove } 1^2 + 2^2 + 3^2 + \cdots + k^2 + (k+1)^2 \\
 = \frac{1}{6}(k+1)((k+1)+1)(2(k+1)+1)
 \end{aligned}$$

$$\begin{aligned}
 \text{LHS} &= 1^2 + 2^2 + 3^2 + \cdots + k^2 + (k+1)^2 \\
 &= \frac{1}{6}k(k+1)(2k+1) + (k+1)^2 \quad \text{by the induction hypothesis (**)} \\
 &= (k+1) \times \left(\frac{1}{6}k(2k+1) + (k+1) \right) \\
 &= \frac{1}{6}(k+1)(k(2k+1) + 6k+6) \\
 &= \frac{1}{6}(k+1)(2k^2 + k + 6k + 6)
 \end{aligned}$$

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$$\begin{aligned}
 &= \frac{1}{6}(k+1)(2k^2+7k+6) \\
 &= \frac{1}{6}(k+1)(k+2)(2k+3) \\
 &= \frac{1}{6}(k+1)((k+1)+1)(2(k+1)+1) \\
 &= \text{RHS}
 \end{aligned}$$

C

It follows from parts **A** and **B** by mathematical induction, that:

$$1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{1}{6}n(n+1)(2n+1) \text{ for all integers } n \geq 1.$$

2g **A**

When $n = 1$,

$$\begin{aligned}
 \text{RHS} &= \frac{1}{3}(1)(2(1)-1)(2(1)+1) \\
 &= \frac{1}{3} \times 1 \times 1 \times 3 \\
 &= 1 \\
 &= \text{LHS}
 \end{aligned}$$

so the statement is true for $n = 1$.

B

Suppose that $k \geq 1$ is a positive integer for which the statement is true.

$$\text{That is, suppose } 1^2 + 3^2 + 5^2 + \cdots + (2k-1)^2 = \frac{1}{3}k(2k-1)(2k+1) \quad (**)$$

We prove the statement for $n = k+1$.

$$\begin{aligned}
 \text{That is, we prove } 1^2 + 3^2 + 5^2 + \cdots + (2k-1)^2 + (2(k+1)-1)^2 \\
 = \frac{1}{3}(k+1)(2(k+1)-1)(2(k+1)+1)
 \end{aligned}$$

$$\begin{aligned}
 \text{LHS} &= 1^2 + 3^2 + 5^2 + \cdots + (2k-1)^2 + (2(k+1)-1)^2 \\
 &= \frac{1}{3}k(2k-1)(2k+1) + (2(k+1)-1)^2 \text{ by the induction hypothesis } (**) \\
 &= \frac{1}{3}k(2k-1)(2k+1) + (2k+2-1)^2 \\
 &= \frac{1}{3}k(2k-1)(2k+1) + (2k+1)^2 \\
 &= (2k+1) \left(\frac{1}{3}k(2k-1) + (2k+1) \right)
 \end{aligned}$$

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$$\begin{aligned}
&= \frac{1}{3}(2k+1)(k(2k-1) + 3(2k+1)) \\
&= \frac{1}{3}(2k+1)(2k^2 - k + 6k + 3) \\
&= \frac{1}{3}(2k+1)(2k^2 + 5k + 3) \\
&= \frac{1}{3}(2k+1)(k+1)(2k+3) \\
&= \frac{1}{3}(k+1)(2k+1)(2k+3) \\
&= \frac{1}{3}(k+1)(2(k+1)-1)(2(k+1)+1) \\
&= \text{RHS}
\end{aligned}$$

C

It follows from parts **A** and **B** by mathematical induction, that:

$$1^2 + 3^2 + 5^2 + \dots + (2n-1)^2 = \frac{1}{3}n(2n-1)(2n+1) \text{ for all integers } n \geq 1.$$

2h **A**

When $n = 1$,

$$\begin{aligned}
\text{RHS} &= \frac{1}{1+1} \\
&= \frac{1}{2} \\
&= \frac{1}{1 \times 2} \\
&= \text{LHS}
\end{aligned}$$

so the statement is true for $n = 1$.

B

Suppose that $k \geq 1$ is a positive integer for which the statement is true.

$$\text{That is, suppose } \frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \dots + \frac{1}{k(k+1)} = \frac{k}{k+1} \quad (**)$$

We prove the statement for $n = k + 1$.

$$\text{That is, we prove } \frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \dots + \frac{1}{k(k+1)} + \frac{1}{(k+1)((k+1)+1)} = \frac{k+1}{(k+1)+1}$$

$$\begin{aligned}
\text{LHS} &= \frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \dots + \frac{1}{k(k+1)} + \frac{1}{(k+1)((k+1)+1)} \\
&= \frac{k}{k+1} + \frac{1}{(k+1)((k+1)+1)} \quad \text{by the induction hypothesis (**)}
\end{aligned}$$

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$$\begin{aligned}
&= \frac{k}{k+1} + \frac{1}{(k+1)(k+2)} \\
&= \frac{k(k+2)}{(k+1)(k+2)} + \frac{1}{(k+1)(k+2)} \\
&= \frac{k^2 + 2k}{(k+1)(k+2)} + \frac{1}{(k+1)(k+2)} \\
&= \frac{k^2 + 2k + 1}{(k+1)(k+2)} \\
&= \frac{(k+1)^2}{(k+1)(k+2)} \\
&= \frac{k+1}{k+2} \\
&= \text{RHS}
\end{aligned}$$

C

It follows from parts **A** and **B** by mathematical induction, that:

$$\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \dots + \frac{1}{n(n+1)} = \frac{n}{n+1} \text{ for all integers } n \geq 1.$$

2i **A**

When $n = 1$,

$$\begin{aligned}
\text{RHS} &= \frac{1}{2(1) + 1} \\
&= \frac{1}{3} \\
&= \frac{1}{1 \times 3} \\
&= \text{LHS}
\end{aligned}$$

so the statement is true for $n = 1$.

B

Suppose that $k \geq 1$ is a positive integer for which the statement is true.

$$\text{That is, suppose } \frac{1}{1 \times 3} + \frac{1}{3 \times 5} + \frac{1}{5 \times 7} + \dots + \frac{1}{(2k-1)(2k+1)} = \frac{k}{2k+1} \quad (**)$$

We prove the statement for $n = k + 1$.

That is, we prove

$$\frac{1}{1 \times 3} + \frac{1}{3 \times 5} + \frac{1}{5 \times 7} + \dots + \frac{1}{(2k-1)(2k+1)} + \frac{1}{(2(k+1)-1)(2(k+1)+1)} = \frac{(k+1)}{2(k+1)+1}$$

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$$\begin{aligned}
\text{LHS} &= \frac{1}{1 \times 3} + \frac{1}{3 \times 5} + \frac{1}{5 \times 7} + \cdots + \frac{1}{(2k-1)(2k+1)} \\
&\quad + \frac{1}{(2(k+1)-1)(2(k+1)+1)} \\
&= \frac{k}{2k+1} + \frac{1}{(2(k+1)-1)(2(k+1)+1)} \quad \text{by the induction hypothesis (**)} \\
&= \frac{k}{2k+1} + \frac{1}{(2k+1)(2k+3)} \\
&= \frac{k(2k+3)}{(2k+1)(2k+3)} + \frac{1}{(2k+1)(2k+3)} \\
&= \frac{2k^2 + 3k}{(2k+1)(2k+3)} + \frac{1}{(2k+1)(2k+3)} \\
&= \frac{2k^2 + 3k + 1}{(2k+1)(2k+3)} \\
&= \frac{(2k+1)(k+1)}{(2k+1)(2k+3)} \\
&= \frac{k+1}{2k+3} \\
&= \frac{k+1}{2(k+1)+1} \\
&= \text{RHS}
\end{aligned}$$

CIt follows from parts **A** and **B** by mathematical induction, that:

$$\frac{1}{1 \times 3} + \frac{1}{3 \times 5} + \frac{1}{5 \times 7} + \cdots + \frac{1}{(2n-1)(2n+1)} = \frac{n}{2n+1} \text{ for all integers } n \geq 1.$$

$$3h \quad \lim_{n \rightarrow \infty} \frac{n}{n+1} = \lim_{n \rightarrow \infty} \frac{1}{1+\frac{1}{n}} = \frac{1}{1+0} = \frac{1}{1} = 1$$

$$3i \quad \lim_{n \rightarrow \infty} \frac{n}{2n+1} = \lim_{n \rightarrow \infty} \frac{1}{2+\frac{1}{n}} = \frac{1}{2+0} = \frac{1}{2}$$

4a AWhen $n = 1$,

$$\begin{aligned}
\text{RHS} &= \frac{1}{12} 1(1+1)(1+2)(3+1) \\
&= \frac{1}{12} (1)(2)(3)(4) \\
&= \frac{1}{12} (24)
\end{aligned}$$

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$$\begin{aligned}
 &= 2 \\
 &= 1^2 \times 2 \\
 &= \text{LHS}
 \end{aligned}$$

so the statement is true for $n = 1$.

B

Suppose that $k \geq 1$ is a positive integer for which the statement is true.

That is, suppose $1^2 \times 2 + 2^2 \times 3 + 3^2 \times 4 + \cdots + k^2(k+1)$

$$= \frac{1}{12}k(k+1)(k+2)(3k+1) \quad (**)$$

We prove the statement for $n = k+1$.

That is, we prove

$$\begin{aligned}
 &1^2 \times 2 + 2^2 \times 3 + 3^2 \times 4 + \cdots + (k)^2(k+1) + (k+1)^2((k+1)+1) \\
 &= \frac{1}{12}(k+1)((k+1)+1)((k+1)+2)(3(k+1)+1)
 \end{aligned}$$

$$\begin{aligned}
 \text{LHS} &= 1^2 \times 2 + 2^2 \times 3 + 3^2 \times 4 + \cdots + (k)^2(k+1) + (k+1)^2((k+1)+1) \\
 &= \frac{1}{12}k(k+1)(k+2)(3k+1) + (k+1)^2((k+1)+1)
 \end{aligned}$$

by the induction hypothesis (**)

$$\begin{aligned}
 &= \frac{1}{12}k(k+1)(k+2)(3k+1) + (k+1)^2(k+2) \\
 &= (k+1)(k+2) \left(\frac{1}{12}k(3k+1) + k+1 \right) \\
 &= \frac{1}{12}(k+1)(k+2)(k(3k+1) + 12k+12) \\
 &= \frac{1}{12}(k+1)(k+2)(3k^2 + k + 12k + 12) \\
 &= \frac{1}{12}(k+1)(k+2)(3k^2 + 13k + 12) \\
 &= \frac{1}{12}(k+1)(k+2)(k+3)(3k+4) \\
 &= \frac{1}{12}(k+1)((k+1)+1)((k+1)+2)(3(k+1)+1) \\
 &= \text{RHS}
 \end{aligned}$$

C

It follows from parts **A** and **B** by mathematical induction, that:

$1^2 \times 2 + 2^2 \times 3 + 3^2 \times 4 + \cdots + n^2(n+1) = \frac{1}{12}n(n+1)(n+2)(3n+1)$ for all integers $n \geq 1$.

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4b A

When $n = 1$,

$$\begin{aligned}
 \text{RHS} &= \frac{1}{12} 1(1+1)(1+2)(3+5) \\
 &= \frac{1}{12} (1)(2)(3)(8) \\
 &= \frac{1}{12} (48) \\
 &= 4 \\
 &= 1 \times 2^2 \\
 &= \text{LHS}
 \end{aligned}$$

so the statement is true for $n = 1$.

B

Suppose that $k \geq 1$ is a positive integer for which the statement is true.That is, suppose $1 \times 2^2 + 2 \times 3^2 + 3 \times 4^2 + \dots + k(k+1)^2$

$$= \frac{1}{12} k(k+1)(k+2)(3k+5) \quad (**)$$

We prove the statement for $n = k + 1$.

That is, we prove

$$\begin{aligned}
 &1 \times 2^2 + 2 \times 3^2 + 3 \times 4^2 + \dots + k(k+1)^2 + (k+1)((k+1)+1)^2 \\
 &= \frac{1}{12} (k+1)((k+1)+1)((k+1)+2)(3(k+1)+5)
 \end{aligned}$$

$$\begin{aligned}
 \text{LHS} &= 1 \times 2^2 + 2 \times 3^2 + 3 \times 4^2 + \dots + k(k+1)^2 + (k+1)((k+1)+1)^2 \\
 &= \frac{1}{12} k(k+1)(k+2)(3k+5) + (k+1)((k+1)+1)^2
 \end{aligned}$$

by the induction hypothesis (**)

$$\begin{aligned}
 &= \frac{1}{12} k(k+1)(k+2)(3k+5) + (k+1)(k+2)^2 \\
 &= (k+1)(k+2) \left(\frac{1}{12} k(3k+5) + (k+2) \right) \\
 &= \frac{1}{12} (k+1)(k+2)(k(3k+5) + 12(k+2)) \\
 &= \frac{1}{12} (k+1)(k+2)(3k^2 + 5k + 12k + 24) \\
 &= \frac{1}{12} (k+1)(k+2)(3k^2 + 17k + 24) \\
 &= \frac{1}{12} (k+1)(k+2)(k+3)(3k+8)
 \end{aligned}$$

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$$\begin{aligned}
 &= \frac{1}{12}(k+1)((k+1)+1)((k+1)+2)(3(k+1)+5) \\
 &= \text{RHS}
 \end{aligned}$$

C

It follows from parts **A** and **B** by mathematical induction, that:

$$1 \times 2^2 + 2 \times 3^2 + 3 \times 4^2 + \cdots + n(n+1)^2 = \frac{1}{12}n(n+1)(n+2)(3n+5) \quad \text{for all integers } n \geq 1.$$

4c **A**

When $n = 1$,

$$\begin{aligned}
 \text{RHS} &= 1 \times 2^1 \\
 &= 2 \\
 &= 2 \times 1 \\
 &= 2 \times 2^0 \\
 &= \text{LHS}
 \end{aligned}$$

so the statement is true for $n = 1$.

B

Suppose that $k \geq 1$ is a positive integer for which the statement is true.

$$\begin{aligned}
 \text{That is, suppose } 2 \times 2^0 + 3 \times 2^1 + 4 \times 2^2 + \cdots + (k+1) \times 2^{k-1} \\
 = k \times 2^k \quad (**)
 \end{aligned}$$

We prove the statement for $n = k + 1$.

That is, we prove

$$\begin{aligned}
 2 \times 2^0 + 3 \times 2^1 + 4 \times 2^2 + \cdots + (k+1) \times 2^{k-1} + ((k+1)+1) \times 2^{(k+1)-1} \\
 = (k+1) \times 2^{(k+1)}
 \end{aligned}$$

$$\begin{aligned}
 \text{LHS} &= 2 \times 2^0 + 3 \times 2^1 + 4 \times 2^2 + \cdots + (k+1) \times 2^{k-1} + ((k+1)+1) \times 2^{(k+1)-1} \\
 &= k \times 2^k + ((k+1)+1) \times 2^{(k+1)-1} \quad \text{by the induction hypothesis (**)} \\
 &= k \times 2^k + ((k+1)+1) \times 2^k \\
 &= k \times 2^k + (k+2) \times 2^k \\
 &= (k+k+2) \times 2^k \\
 &= (2k+2) \times 2^k \\
 &= (k+1) \times 2 \times 2^k \\
 &= (k+1) \times 2^{k+1} \\
 &= \text{RHS}
 \end{aligned}$$

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CIt follows from parts **A** and **B** by mathematical induction, that:

$$2 \times 2^0 + 3 \times 2^1 + 4 \times 2^2 + \cdots + (n+1) \times 2^{n-1} = n \times 2^n \text{ for all integers } n \geq 1.$$

5a **A**When $n = 1$,

$$\begin{aligned} \text{RHS} &= (1+1)! - 1 \\ &= 2! - 1 \\ &= 2 - 1 \\ &= 1 \\ &= 1 \times 1! \\ &= \text{LHS} \end{aligned}$$

so the statement is true for $n = 1$.**B**Suppose that $k \geq 1$ is a positive integer for which the statement is true.

$$\text{That is, suppose } 1 \times 1! + 2 \times 2! + 3 \times 3! + \cdots + k \times k! = (k+1)! - 1 \quad (**)$$

We prove the statement for $n = k+1$.

$$\begin{aligned} \text{That is, we prove } 1 \times 1! + 2 \times 2! + 3 \times 3! + \cdots + k \times k! + (k+1) \times (k+1)! \\ = (k+2)! - 1 \end{aligned}$$

$$\begin{aligned} \text{LHS} &= 1 \times 1! + 2 \times 2! + 3 \times 3! + \cdots + k \times k! + (k+1) \times (k+1)! \\ &= (k+1)! - 1 + (k+1) \times (k+1)! \quad \text{by the induction hypothesis (**)} \\ &= (k+1)! + (k+1) \times (k+1)! - 1 \\ &= (k+1)! (1 + k+1) - 1 \\ &= (k+1)! (k+2) - 1 \\ &= (k+2)! - 1 \\ &= \text{RHS} \end{aligned}$$

CIt follows from parts **A** and **B** by mathematical induction, that:

$$1 \times 1! + 2 \times 2! + 3 \times 3! + \cdots + n \times n! = (n+1)! - 1 \text{ for all integers } n \geq 1.$$

5b **A**When $n = 1$,

$$\begin{aligned} \text{RHS} &= 1(1+1)! \\ &= 2! \end{aligned}$$

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$$\begin{aligned}
 &= 2 \\
 &= 2 \times 1! \\
 &= \text{LHS}
 \end{aligned}$$

so the statement is true for $n = 1$.

B

Suppose that $k \geq 1$ is a positive integer for which the statement is true.

That is, suppose $2 \times 1! + 5 \times 2! + 10 \times 3! + \cdots + (k^2 + 1)k! = k(k + 1)! \quad (**)$

We prove the statement for $n = k + 1$.

That is, we prove

$$\begin{aligned}
 2 \times 1! + 5 \times 2! + 10 \times 3! + \cdots + (k^2 + 1)k! + ((k + 1)^2 + 1)(k + 1)! \\
 = (k + 1)(k + 2)!
 \end{aligned}$$

$$\begin{aligned}
 \text{LHS} &= 2 \times 1! + 5 \times 2! + 10 \times 3! + \cdots + (k^2 + 1)k! + ((k + 1)^2 + 1)(k + 1)! \\
 &= k(k + 1)! + ((k + 1)^2 + 1)(k + 1)! \quad \text{by the induction hypothesis (**)} \\
 &= (k + 1)! (k + ((k + 1)^2 + 1)) \\
 &= (k + 1)! (k + (k^2 + 2k + 1 + 1)) \\
 &= (k + 1)! (k^2 + 3k + 2) \\
 &= (k + 1)! (k + 1)(k + 2) \\
 &= (k + 1)(k + 2)(k + 1)! \\
 &= (k + 1)(k + 2)! \\
 &= \text{RHS}
 \end{aligned}$$

C

It follows from parts **A** and **B** by mathematical induction, that:

$$2 \times 1! + 5 \times 2! + 10 \times 3! + \cdots + (n^2 + 1)n! = n(n + 1)! \text{ for all integers } n \geq 1.$$

5c **A**

When $n = 1$,

$$\begin{aligned}
 \text{RHS} &= 1 - \frac{1}{(1 + 1)!} \\
 &= 1 - \frac{1}{2!} \\
 &= 1 - \frac{1}{2} \\
 &= \frac{1}{2} \\
 &= \frac{1}{2!}
 \end{aligned}$$

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$$= \text{LHS}$$

so the statement is true for $n = 1$.

B

Suppose that $k \geq 1$ is a positive integer for which the statement is true.

$$\text{That is, suppose } \frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \cdots + \frac{k}{(k+1)!} = 1 - \frac{1}{(k+1)!} \quad (**)$$

We prove the statement for $n = k + 1$.

$$\text{That is, we prove } \frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \cdots + \frac{k}{(k+1)!} + \frac{k+1}{(k+2)!} = 1 - \frac{1}{(k+2)!}$$

$$\begin{aligned} \text{LHS} &= \frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \cdots + \frac{k}{(k+1)!} + \frac{k+1}{(k+2)!} \\ &= 1 - \frac{1}{(k+1)!} + \frac{k+1}{(k+2)!} \quad \text{by the induction hypothesis (**)} \\ &= 1 - \frac{k+2}{(k+2)(k+1)!} + \frac{k+1}{(k+2)!} \\ &= 1 - \frac{k+2}{(k+2)!} + \frac{k+1}{(k+2)!} \\ &= 1 - \frac{1}{(k+2)!} \\ &= \text{RHS} \end{aligned}$$

C

It follows from parts **A** and **B** by mathematical induction, that:

$$\frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \cdots + \frac{n}{(n+1)!} = 1 - \frac{1}{(n+1)!} \text{ for all integers } n \geq 1.$$

6a Suppose that $k \geq 1$ is a positive integer for which the statement is true.

$$\text{That is, suppose } 1 + 3 + 5 + \cdots + (2k - 1) = (k)^2 + 2 \quad (**)$$

We prove the statement for $n = k + 1$.

$$\text{That is, we prove } 1 + 3 + 5 + \cdots + (2k - 1) + (2(k + 1) - 1) = (k + 1)^2 + 2$$

$$\begin{aligned} \text{LHS} &= 1 + 3 + 5 + \cdots + (2k - 1) + (2(k + 1) - 1) \\ &= k^2 + 2 + (2(k + 1) - 1) \quad \text{by the induction hypothesis (**)} \\ &= k^2 + 2 + (2k + 2 - 1) \\ &= k^2 + 2 + 2k + 1 \\ &= k^2 + 2k + 3 \\ &= k^2 + 2k + 1 + 2 \\ &= (k + 1)^2 + 2 \\ &= \text{RHS} \end{aligned}$$

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6b When $n = 1$,

$$\begin{aligned}\text{RHS} &= 1^2 + 2 \\ &= 3 \\ &\neq \text{LHS}\end{aligned}$$

so the statement is **not** true for $n = 1$.7a **A**When $n = 1$,

$$\begin{aligned}\text{RHS} &= 1(1 + 1) + 1 \\ &= 2 + 1 \\ &= 3 \\ &= \text{LHS}\end{aligned}$$

so the statement is true for $n = 1$.**B**Suppose that $k \geq 1$ is a positive integer for which the statement is true.That is, suppose $3 + 6 + 9 + \cdots + 3k = k(k + 1) + 1$ (**)We prove the statement for $n = k + 1$.That is, we prove $3 + 6 + 9 + \cdots + 3k + 3(k + 1) = (k + 1)((k + 1) + 1) + 1$

$$\begin{aligned}\text{LHS} &= 3 + 6 + 9 + \cdots + 3k + 3(k + 1) \\ &= k(k + 1) + 1 + 3(k + 1) \quad \text{by the induction hypothesis (**)} \\ &= k(k + 1) + 3(k + 1) + 1 \\ &= (k + 3)(k + 1) + 1 \\ &= k^2 + 4k + 3 + 1 \\ &= k^2 + 4k + 4\end{aligned}$$

But:

$$\begin{aligned}\text{RHS} &= (k + 1)((k + 1) + 1) + 1 \\ &= (k + 1)(k + 2) + 1 \\ &= k^2 + 3k + 2 + 1 \\ &= k^2 + 3k + 3 \\ &\neq \text{LHS}\end{aligned}$$

So the proof breaks down.

7b If it is true for $n = k$, it does not follow that it is true for $n = k + 1$.

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8a $P(-1) = 4(-1)^3 + 18(-1)^2 + 23(-1) + 9 = 0$, hence $n + 1$ is a factor.

$$\begin{aligned} P(n) &= 4n^3 + 18n^2 + 23n + 9 \\ &= (n + 1)(4n^2 + 14n + 9) \end{aligned}$$

8b **A**

When $n = 1$,

$$\begin{aligned} \text{RHS} &= \frac{1}{3}(1)(4 + 6 - 1) \\ &= \frac{9}{3} \\ &= 3 \\ &= 1 \times 3 \\ &= \text{LHS} \end{aligned}$$

so the statement is true for $n = 1$.

B

Suppose that $k \geq 1$ is a positive integer for which the statement is true.

That is, suppose $1 \times 3 + 3 \times 5 + 5 \times 7 + \cdots + (2k - 1)(2k + 1)$

$$= \frac{1}{3}k(4k^2 + 6k - 1) \quad (**)$$

We prove the statement for $n = k + 1$.

That is, we prove

$$\begin{aligned} 1 \times 3 + 3 \times 5 + 5 \times 7 + \cdots + (2k - 1)(2k + 1) + (2(k + 1) - 1)(2(k + 1) + 1) \\ = \frac{1}{3}(k + 1)(4(k + 1)^2 + 6(k + 1) - 1) \end{aligned}$$

$$\begin{aligned} \text{LHS} &= 1 \times 3 + 3 \times 5 + 5 \times 7 + \cdots + (2k - 1)(2k + 1) \\ &\quad + (2(k + 1) - 1)(2(k + 1) + 1) \\ &= \frac{1}{3}k(4k^2 + 6k - 1) + (2(k + 1) - 1)(2(k + 1) + 1) \end{aligned}$$

by the induction hypothesis (**)

$$\begin{aligned} &= \frac{1}{3}k(4k^2 + 6k - 1) + (2k + 2 - 1)(2k + 2 + 1) \\ &= \frac{1}{3}k(4k^2 + 6k - 1) + (2k + 1)(2k + 3) \\ &= \frac{1}{3}(k(4k^2 + 6k - 1) + 3(2k + 1)(2k + 3)) \\ &= \frac{1}{3}(k(4k^2 + 6k - 1) + 3(4k^2 + 8k + 3)) \end{aligned}$$

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$$\begin{aligned}
&= \frac{1}{3}(4k^3 + 6k^2 - k + 12k^2 + 24k + 9) \\
&= \frac{1}{3}(4k^3 + 18k^2 + 23k + 9) \\
&= \frac{1}{3}(k+1)(4k^2 + 14k + 9) \quad \text{using answer to question 8a} \\
&= \frac{1}{3}(k+1)(4k^2 + 8k + 4 + 6k + 6 - 1) \\
&= \frac{1}{3}(k+1)(4(k^2 + 2k + 1) + 6(k+1) - 1) \\
&= \frac{1}{3}(k+1)(4(k+1)^2 + 6(k+1) - 1) \\
&= \text{RHS}
\end{aligned}$$

C

It follows from parts **A** and **B** by mathematical induction, that:

$1 \times 3 + 3 \times 5 + 5 \times 7 + \cdots + (2n-1)(2n+1) = \frac{1}{3}n(4n^2 + 6n - 1)$ for all integers $n \geq 1$.

9 **A**

When $n = 1$,

$$\begin{aligned}
\text{RHS} &= \frac{1}{6}(1)(1+1)(1+2) \\
&= \frac{1}{6}(1)(2)(3) \\
&= 1 \\
&= \text{LHS}
\end{aligned}$$

so the statement is true for $n = 1$.

B

Suppose that $k \geq 1$ is a positive integer for which the statement is true.

That is, suppose $1 + (1+2) + (1+2+3) + \cdots + (1+2+3+\cdots+k)$

$$= \frac{1}{6}k(k+1)(k+2) \quad (**)$$

We prove the statement for $n = k+1$.

That is, we prove $1 + (1+2) + (1+2+3) + \cdots + (1+2+3+\cdots+k) + (1+2+3+\cdots+k+(k+1)) = \frac{1}{6}(k+1)((k+1)+1)((k+1)+2)$

$$\begin{aligned}
\text{LHS} &= 1 + (1+2) + (1+2+3) + \cdots + (1+2+3+\cdots+k) \\
&\quad + (1+2+3+\cdots+k+(k+1))
\end{aligned}$$

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$$= \frac{1}{6}k(k+1)(k+2) + (1+2+3+\dots+k+(k+1))$$

by the induction hypothesis (**)

$$= \frac{1}{6}k(k+1)(k+2) + (1+2+3+\dots+k) + (k+1)$$

$$= \frac{1}{6}k(k+1)(k+2) + \frac{1}{2}k(k+1) + (k+1) \quad \text{from Question 2a}$$

$$= (k+1) \left(\frac{1}{6}(k)(k+2) + \frac{1}{2}k+1 \right)$$

$$= \frac{1}{6}(k+1)((k)(k+2) + 3k+6)$$

$$= \frac{1}{6}(k+1)(k^2+2k+3k+6)$$

$$= \frac{1}{6}(k+1)(k^2+5k+6)$$

$$= \frac{1}{6}(k+1)(k+2)(k+3)$$

$$= \frac{1}{6}(k+1)((k+1)+1)((k+1)+2)$$

$$= \text{RHS}$$

CIt follows from parts **A** and **B** by mathematical induction, that:

$1 + (1+2) + (1+2+3) + \dots + (1+2+3+\dots+n) = \frac{1}{6}n(n+1)(n+2)$ for all integers $n \geq 1$.

10 AWhen $n = 1$,

$$\text{RHS} = 2^1(1)$$

$$= 2$$

$$= (1+1)$$

$$= \text{LHS}$$

so the statement is true for $n = 1$.**B**Suppose that $k \geq 1$ is a positive integer for which the statement is true.That is, suppose $(k+1)(k+2)(k+3) \times \dots \times 2k$

$$= 2^k(1 \times 3 \times 5 \times \dots \times (2k-1)) \quad (**)$$

We prove the statement for $n = k+1$.

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$$\begin{aligned}\text{That is, we prove } & ((k+1)+1)((k+1)+2)((k+1)+3) \times \dots \times 2(k+1) \\ & = 2^{k+1}(1 \times 3 \times 5 \times \dots \times (2k-1)(2(k+1)-1)) \quad (**)\end{aligned}$$

$$\begin{aligned}\text{LHS} &= ((k+1)+1)((k+1)+2)((k+1)+3) \times \dots \times 2k \times (2k+1) \times 2(k+1) \\ &= (k+2)(k+3) \times \dots \times 2k \times (2k+1) \times 2(k+1) \\ &= 2(k+1)(k+2)(k+3) \times \dots \times 2k \times (2k+1) \\ &= 2 \times 2^k(1 \times 3 \times 5 \times \dots \times (2k-1)) \times (2k+1)\end{aligned}$$

$$\begin{aligned}& \text{by the induction hypothesis (**),} \\ &= 2^{k+1}(1 \times 3 \times 5 \times \dots \times (2k-1) \times (2k+1)) \\ &= 2^{k+1}(1 \times 3 \times 5 \times \dots \times (2k-1) \times (2(k+1)-1)) \\ &= \text{RHS}\end{aligned}$$

CIt follows from parts **A** and **B** by mathematical induction, that:

$$(n+1)(n+2)(n+3) \times \dots \times 2n = 2^n(1 \times 3 \times 5 \times \dots \times (2n-1)) \text{ for all integers } n \geq 1.$$

11a **A**When $n = 1$,

$$\begin{aligned}\text{RHS} &= \frac{1}{4}(1-1)(1)(1+1)(1+2) \\ &= 0\end{aligned}$$

$$\begin{aligned}\text{LHS} &= \sum_{r=1}^1 (r^3 - r) \\ &= 1^3 - 1 \\ &= 0 \\ &= \text{RHS}\end{aligned}$$

so the statement is true for $n = 1$.**B**Suppose that $k \geq 1$ is a positive integer for which the statement is true.

$$\text{That is, suppose } \sum_{r=1}^k (r^3 - r) = \frac{1}{4}(k-1)(k)(k+1)(k+2) \quad (**)$$

We prove the statement for $n = k + 1$.

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That is, we prove $\sum_{r=1}^{k+1} (r^3 - r)$

$$= \frac{1}{4}((k+1)-1)((k+1))((k+1)+1)((k+1)+2)$$

$$\text{LHS} = \sum_{r=1}^{k+1} (r^3 - r)$$

$$= (k+1)^3 - (k+1) + \sum_{r=1}^k (r^3 - r)$$

$$= (k+1)^3 - (k+1) + \frac{1}{4}(k-1)(k)(k+1)(k+2)$$

by the induction hypothesis (**),

$$= (k+1) \left[(k+1)^2 - 1 + \frac{1}{4}(k-1)(k)(k+2) \right]$$

$$= \frac{1}{4}(k+1)[4(k+1)^2 - 4 + (k-1)(k)(k+2)]$$

$$= \frac{1}{4}(k+1)[4(k+1)^2 - 4 + (k^2 - k)(k+2)]$$

$$= \frac{1}{4}(k+1)[4(k^2 + 2k + 1) - 4 + k^3 + k^2 - 2k]$$

$$= \frac{1}{4}(k+1)[k^3 + 5k^2 + 6k]$$

$$= \frac{1}{4}(k+1)k(k^2 + 5k + 6)$$

$$= \frac{1}{4}(k+1)k(k+2)(k+3)$$

$$= \frac{1}{4}k(k+1)(k+2)(k+3)$$

$$= \frac{1}{4}((k+1)-1)((k+1))((k+1)+1)((k+1)+2)$$

$$= \text{RHS}$$

C

It follows from parts **A** and **B** by mathematical induction, that:

$$\sum_{r=1}^n (r^3 - r) = \frac{1}{4}(n-1)(n)(n+1)(n+2) \text{ for all integers } n \geq 1.$$

11b A

When $n = 1$,

$$\begin{aligned} \text{RHS} &= \frac{1}{2}(1)^3(1+1)^3 \\ &= \frac{8}{2} \end{aligned}$$

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$$= 4$$

$$\begin{aligned}\text{LHS} &= \sum_{r=1}^1 (3r^5 + r^3) \\ &= 3 + 1 \\ &= 4 \\ &= \text{RHS}\end{aligned}$$

so the statement is true for $n = 1$.

B

Suppose that $k \geq 1$ is a positive integer for which the statement is true.

$$\text{That is, suppose } \sum_{r=1}^k (3r^5 + r^3) = \frac{1}{2}k^3(k+1)^3 \quad (**)$$

We prove the statement for $n = k + 1$.

$$\text{That is, we prove } \sum_{r=1}^{k+1} (3r^5 + r^3) = \frac{1}{2}(k+1)^3(k+2)^3$$

$$\begin{aligned}\text{LHS} &= \sum_{r=1}^{k+1} (3r^5 + r^3) \\ &= 3(k+1)^5 + (k+1)^3 + \sum_{r=1}^k (3r^5 + r^3) \\ &= 3(k+1)^5 + (k+1)^3 + \frac{1}{2}k^3(k+1)^3 \quad \text{by the induction hypothesis (**)} \\ &= (k+1)^3 \left[3(k+1)^2 + 1 + \frac{1}{2}k^3 \right] \\ &= \frac{1}{2}(k+1)^3 [6(k^2 + 2k + 1) + 2 + k^3] \\ &= \frac{1}{2}(k+1)^3 [k^3 + 6k^2 + 12k + 8] \\ &= \frac{1}{2}(k+1)^3(k+2)^3 \\ &= \text{RHS}\end{aligned}$$

C

It follows from parts **A** and **B** by mathematical induction, that:

$$\sum_{r=1}^n (3r^5 + r^3) = \frac{1}{2}n^3(n+1)^3 \text{ for all integers } n \geq 1.$$

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11c **A**When $n = 1$,

$$\begin{aligned}\text{RHS} &= (1 - 2 + 3) \times 2^{1+1} - 6 \\ &= 2 \times 4 - 6 \\ &= 2\end{aligned}$$

$$\begin{aligned}\text{LHS} &= \sum_{r=1}^1 r^2 \times 2^r \\ &= 1 \times 2 \\ &= 2 \\ &= \text{RHS}\end{aligned}$$

so the statement is true for $n = 1$.**B**Suppose that $k \geq 1$ is a positive integer for which the statement is true.

$$\text{That is, suppose } \sum_{r=1}^k r^2 \times 2^r = (k^2 - 2k + 3) \times 2^{k+1} - 6 \quad (**)$$

We prove the statement for $n = k + 1$.

$$\text{That is, we prove } \sum_{r=1}^{k+1} r^2 \times 2^r = ((k+1)^2 - 2(k+1) + 3) \times 2^{(k+1)+1} - 6$$

$$\begin{aligned}\text{LHS} &= \sum_{r=1}^{k+1} r^2 \times 2^r \\ &= (k+1)^2 \times 2^{k+1} + \sum_{r=1}^k r^2 \times 2^r \\ &= (k+1)^2 \times 2^{k+1} + (k^2 - 2k + 3) \times 2^{k+1} - 6\end{aligned}$$

by the induction hypothesis (**),

$$\begin{aligned}&= 2^{k+1}[(k+1)^2 + (k^2 - 2k + 3)] - 6 \\ &= 2^{k+1}[k^2 + 2k + 1 + k^2 - 2k + 3] - 6 \\ &= 2^{k+1}[2k^2 + 4] - 6 \\ &= 2(k^2 + 2) \times 2^k - 6 \\ &= (k^2 + 2) \times 2^{k+2} - 6 \\ &= (k^2 + 2k + 1 - 2k - 2 + 3) \times 2^{k+2} - 6 \\ &= ((k+1)^2 - 2(k+1) + 3) \times 2^{(k+1)+1} - 6 \\ &= \text{RHS}\end{aligned}$$

CIt follows from parts **A** and **B** by mathematical induction, that:

$$\sum_{r=1}^n r^2 \times 2^r = (n^2 - 2n + 3) \times 2^{n+1} - 6 \text{ for all integers } n \geq 1.$$

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12 **A**When $n = 1$,

$$\begin{aligned}
 \text{RHS} &= 1 \times H(1) \\
 &= 1 \times 1 \\
 &= 1 \\
 &= \text{LHS}
 \end{aligned}$$

so the statement is true for $n = 1$.**B**Suppose that $k \geq 1$ is a positive integer for which the statement is true.

$$\text{That is, suppose } k + H(1) + H(2) + \cdots + H(k-1) = k \times H(k) \quad (**)$$

We prove the statement for $n = k + 1$.

$$\begin{aligned}
 \text{That is, we prove } (k+1) + H(1) + H(2) + \cdots + H(k-1) + H(k) \\
 = (k+1) \times H(k+1)
 \end{aligned}$$

$$\begin{aligned}
 \text{LHS} &= (k+1) + H(1) + H(2) + \cdots + H(k-1) + H(k) \\
 &= (k + H(1) + H(2) + \cdots + H(k-1)) + (1 + H(k)) \\
 &= k \times H(k) + (1 + H(k)) \quad \text{by the induction hypothesis } (**) \\
 &= k \times H(k) + 1 + H(k) \\
 &= 1 + (k+1)H(k) \\
 &= 1 + (k+1) \left(1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{k} \right) \\
 &= (k+1) \left(1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{k} + \frac{1}{k+1} \right) \\
 &= (k+1) \times H(k+1) \\
 &= \text{RHS}
 \end{aligned}$$

CIt follows from parts **A** and **B** by mathematical induction, that:

$$n + H(1) + H(2) + \cdots + H(n-1) = n \times H(n) \text{ for all integers } n \geq 1.$$

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13a

$$\begin{aligned}
 \text{LHS} &= \frac{\cos \alpha - \cos \alpha \cos 2\beta + \sin \alpha \sin 2\beta}{2 \sin \beta} \\
 &= \frac{\cos \alpha - \cos \alpha (1 - 2 \sin^2 \beta) + \sin \alpha \times 2 \sin \beta \cos \beta}{2 \sin \beta} \\
 &= \frac{2 \cos \alpha \sin^2 \beta + 2 \sin \alpha \sin \beta \cos \beta}{2 \sin \beta} \\
 &= \cos \alpha \sin \beta + \sin \alpha \cos \beta \\
 &= \sin(\alpha + \beta) \\
 &= \text{RHS}
 \end{aligned}$$

13b A

When $n = 1$,

$$\begin{aligned}
 \text{RHS} &= \frac{1 - \cos 2\theta}{2 \sin \theta} \\
 &= \frac{1 - (1 - 2 \sin^2 \theta)}{2 \sin \theta} \\
 &= \frac{2 \sin^2 \theta}{2 \sin \theta} \\
 &= \sin \theta \\
 &= \text{LHS}
 \end{aligned}$$

so the statement is true for $n = 1$.**B**Suppose that $k \geq 1$ is a positive integer for which the statement is true.

$$\text{That is, suppose } \sin \theta + \sin 3\theta + \cdots + \sin(2k-1)\theta = \frac{1 - \cos 2k\theta}{2 \sin \theta} \quad (**)$$

We prove the statement for $n = k + 1$.

That is, we prove

$$\sin \theta + \sin 3\theta + \cdots + \sin(2k-1)\theta + \sin(2k+1)\theta = \frac{1 - \cos 2(k+1)\theta}{2 \sin \theta}$$

$$\begin{aligned}
 \text{LHS} &= \sin \theta + \sin 3\theta + \cdots + \sin(2k-1)\theta + \sin(2k+1)\theta \\
 &= \frac{1 - \cos 2k\theta}{2 \sin \theta} + \sin(2k+1)\theta && \text{by the induction hypothesis (**),} \\
 &= \frac{1 - \cos 2k\theta + 2 \sin \theta \sin(2k+1)\theta}{2 \sin \theta} \\
 &= \frac{1 - \cos 2k\theta + (\cos 2k\theta - \cos(2k\theta + 2\theta))}{2 \sin \theta} && \text{using part a with } \alpha = 2k\theta \text{ and } \beta = \theta
 \end{aligned}$$

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$$\begin{aligned} &= \frac{1 - \cos(2k\theta + 2\theta)}{2 \sin \theta} \\ &= \frac{1 - \cos(2(k+1)\theta)}{2 \sin \theta} \\ &= \text{RHS} \end{aligned}$$

C

It follows from parts **A** and **B** by mathematical induction, that:

$$\sin \theta + \sin 3\theta + \cdots + \sin(2n-1)\theta = \frac{1 - \cos 2n\theta}{2 \sin \theta} \text{ for all integers } n \geq 1.$$

Solutions to Exercise 2B

1 **A**

When $n = 1$, $7^n - 1 = 7^1 - 1 = 7 - 1 = 6$ which is divisible by 6

so the statement is true for $n = 1$.

B

Suppose that $k \geq 1$ is a positive integer for which the statement is true.

That is, suppose $7^k - 1 = 6m$, for some integer m . (**)

We prove the statement for $n = k + 1$.

That is, we prove $7^{k+1} - 1$ is divisible by 6.

$$\begin{aligned}
 7^{k+1} - 1 &= 7 \times 7^k - 1 \\
 &= 7 \times 7^k - 7 + 6 \\
 &= 7(7^k - 1) + 6 \\
 &= 7(6m) + 6 && \text{by the induction hypothesis (**)} \\
 &= 6(7m + 1), \text{ which is divisible by 6 as required.}
 \end{aligned}$$

C

It follows from parts **A** and **B** by mathematical induction that the statement is true for all positive integers n .

2a **A**

When $n = 1$, $5^n - 1 = 5^1 - 1 = 5 - 1 = 4$ which is divisible by 4

so the statement is true for $n = 1$.

B

Suppose that $k \geq 1$ is a positive integer for which the statement is true.

That is, suppose $5^k - 1 = 4m$, for some integer m . (**)

We prove the statement for $n = k + 1$.

That is, we prove $5^{k+1} - 1$ is divisible by 4.

$$\begin{aligned}
 5^{k+1} - 1 &= 5 \times 5^k - 1 \\
 &= 5 \times 5^k - 5 + 4 \\
 &= 5(5^k - 1) + 4 \\
 &= 5(4m) + 4 && \text{by the induction hypothesis (**)} \\
 &= 4(5m + 1), \text{ which is divisible by 4 as required.}
 \end{aligned}$$

Chapter 2 worked solutions – Mathematical induction

C

It follows from parts **A** and **B** by mathematical induction that the statement is true for all integers $n \geq 1$.

2b A

When $n = 1$, $9^n + 3 = 9^1 + 3 = 12 = 2 \times 6$ which is divisible by 6

so the statement is true for $n = 1$.

B

Suppose that $k \geq 1$ is a positive integer for which the statement is true.

That is, suppose $9^k + 3 = 6m$, for some integer m . (**)

We prove the statement for $n = k + 1$.

That is, we prove $9^{k+1} + 3$ is divisible by 6.

$$\begin{aligned} 9^{k+1} + 3 &= 9 \times 9^k + 3 \\ &= 9(9^k + 3) - 9 \times 3 + 3 \\ &= 9(9^k + 3) - 27 + 3 \\ &= 9(6m) - 24 \quad \text{by the induction hypothesis (**)} \\ &= 6(9m - 4), \text{ which is divisible by 6 as required.} \end{aligned}$$

C

It follows from parts **A** and **B** by mathematical induction that the statement is true for all integers $n \geq 1$.

2c A

When $n = 1$, $3^2 + 7 = 9 + 7 = 16 = 8 \times 2$ which is divisible by 8

so the statement is true for $n = 1$.

B

Suppose that $k \geq 1$ is a positive integer for which the statement is true.

That is, suppose $3^{2k} + 7 = 8m$, for some integer m . (**)

We prove the statement for $n = k + 1$.

That is, we prove $3^{2(k+1)} + 7$ is divisible by 8.

$$\begin{aligned} 3^{2(k+1)} + 7 &= 3^{2k+2} + 7 \\ &= 3^2 \times 3^{2k} + 7 \end{aligned}$$

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$$\begin{aligned}
 &= 9 \times 3^{2k} + 7 \\
 &= 9 \times (3^{2k} + 7) - 7 \times 9 + 7 \\
 &= 9 \times (8m) - 56 \quad \text{by the induction hypothesis (**)} \\
 &= 8 \times 9m - 8 \times 7 \\
 &= 8(9m - 7), \text{ which is divisible by 8 as required.}
 \end{aligned}$$

C

It follows from parts **A** and **B** by mathematical induction that the statement is true for all integers $n \geq 1$.

2d **A**

When $n = 1$, $5^2 - 1 = 25 - 1 = 24$ which is divisible by 24

so the statement is true for $n = 1$.

B

Suppose that $k \geq 1$ is a positive integer for which the statement is true.

That is, suppose $5^{2k} - 1 = 24m$, for some integer m . (**)

We prove the statement for $n = k + 1$.

That is, we prove $5^{2(k+1)} - 1$ is divisible by 24.

$$\begin{aligned}
 5^{2(k+1)} - 1 &= 5^{2k+2} - 1 \\
 &= 5^2 \times 5^{2k} - 1 \\
 &= 25 \times 5^{2k} - 1 \\
 &= 25 \times (5^{2k} - 1) + 25 - 1 \\
 &= 25 \times (24m) + 24 \quad \text{by the induction hypothesis (**)} \\
 &= 24(25m + 1), \text{ which is divisible by 24 as required.}
 \end{aligned}$$

C

It follows from parts **A** and **B** by mathematical induction that the statement is true for all integers $n \geq 1$.

3a

n	0	1	2	3	4
$11^n - 1$	0	10	120	1330	14 640

From this we can hypothesise that the expression will always be divisible by 10.

Chapter 2 worked solutions – Mathematical induction

3b **A**

When $n = 1$, $11^0 - 1 = 1 - 1 = 0$ which is divisible by 10

so the statement is true for $n = 1$.

B

Suppose that $k \geq 1$ is a positive integer for which the statement is true.

That is, suppose, $11^k - 1 = 10m$, for some integer m . (**)

We prove the statement for $n = k + 1$.

That is, we prove $11^{k+1} - 1$ is divisible by 10.

$$\begin{aligned} 11^{k+1} - 1 &= 11 \times 11^k - 1 \\ &= 11(11^k - 1) + 11 - 1 \\ &= 11(10m) + 10 && \text{by the induction hypothesis (**)} \\ &= 10(11m + 1), \text{ which is divisible by 10 as required.} \end{aligned}$$

C

It follows from parts **A** and **B** by mathematical induction that the statement is true for all integers $n \geq 0$.

4a **A**

When $n = 0$, $0^3 + 2(0) = 0 + 0 = 0$ which is divisible by 3

so the statement is true for $n = 0$.

B

Suppose that $k \geq 0$ is an integer for which the statement is true.

That is, suppose $k^3 + 2k = 3m$, for some integer m . (**)

We prove the statement for $n = k + 1$.

That is, we prove $(k + 1)^3 + 2(k + 1)$ is divisible by 3.

$$\begin{aligned} &(k + 1)^3 + 2(k + 1) \\ &= (k + 1)((k + 1)^2 + 2) \\ &= (k + 1)(k^2 + 2k + 1 + 2) \\ &= (k + 1)(k^2 + 2k + 3) \\ &= k^3 + 2k^2 + 3k + k^2 + 2k + 3 \\ &= k^3 + 3k^2 + 5k + 3 \\ &= (k^3 + 2k) + 3k^2 + 3k + 3 \end{aligned}$$

Chapter 2 worked solutions – Mathematical induction

$$\begin{aligned}
 &= 3m + 3k^2 + 3k + 3 && \text{by the induction hypothesis (**)} \\
 &= 3(m + k^2 + k + 1) \text{ which is divisible by 3.}
 \end{aligned}$$

C

It follows from parts **A** and **B** by mathematical induction that the statement is true for all integers $n \geq 0$.

4b **A**

When $n = 0$, $8^0 - 7(0) + 6 = 1 + 0 + 6 = 7$ which is divisible by 7
so the statement is true for $n = 0$.

B

Suppose that $k \geq 0$ is an integer for which the statement is true.

That is suppose, $8^k - 7k + 6 = 7m$, for some integer m . (**)

We prove the statement for $n = k + 1$.

That is, we prove $8^{k+1} - 7(k + 1) + 6$ is divisible by 7.

$$\begin{aligned}
 &8^{k+1} - 7(k + 1) + 6 \\
 &= 8 \times 8^k - 7k - 7 + 6 \\
 &= 8 \times 8^k - 7k - 1 \\
 &= 8(8^k - 7k + 6) + 56k - 48 - 7k - 1 \\
 &= 8(8^k - 7k + 6) + 49k - 49 \\
 &= 8(7m) + 7(7k - 7) && \text{by the induction hypothesis (**)} \\
 &= 7(8m + 7k - 7) \text{ which is divisible by 7.}
 \end{aligned}$$

C

It follows from parts **A** and **B** by mathematical induction that the statement is true for all integers $n \geq 0$.

4c **A**

When $n = 0$, $9(9^0 - 1) - 8(0) = 9 \times 0 = 0$ which is divisible by 64
so the statement is true for $n = 0$.

B

Suppose that $k \geq 0$ is an integer for which the statement is true.

That is, suppose $9(9^k - 1) - 8k = 64m$, for some integer m .

Note that rearranging this gives $9(9^k - 1) = 64m + 8k$. (**)

Chapter 2 worked solutions – Mathematical induction

We prove the statement for $n = k + 1$.

That is, we prove $9(9^{k+1} - 1) - 8(k + 1)$ is divisible by 64.

$$\begin{aligned}
 & 9(9^{k+1} - 1) - 8(k + 1) \\
 &= 9(9 \times 9^k - 1) - 8(k + 1) \\
 &= 81 \times 9^k - 9 - 8k - 8 \\
 &= 81 \times 9^k - 17 - 8k \\
 &= 81 \times 9^k - 81 + 64 - 8k \\
 &= 81(9^k - 1) + 64 - 8k \\
 &= 9 \times 9(9^k - 1) + 64 - 8k \\
 &= 9(64m + 8k) + 64 - 8k && \text{by the induction hypothesis (**)} \\
 &= 9 \times 64m + 9 \times 8k + 64 - 8k \\
 &= 9 \times 64m + 8 \times 8k + 64 \\
 &= 9 \times 64m + 64k + 64 \\
 &= 64(9m + 1k + 1) \text{ which is divisible by 64}
 \end{aligned}$$

C

It follows from parts **A** and **B** by mathematical induction that the statement is true for all integers $n \geq 0$.

5a **A**

When $n = 0$, $5^0 + 2 \times 11^0 = 1 + 2 = 3$ which is divisible by 3

so the statement is true for $n = 0$.

B

Suppose that $k \geq 0$ is an integer for which the statement is true.

That is, suppose $5^k + 2 \times 11^k = 3m$, for some integer m .

Note that rearranging this gives $5^k = 3m - 2 \times 11^k$. (**)

We prove the statement for $n = k + 1$.

That is, we prove $5^{k+1} + 2 \times 11^{k+1}$ is divisible by 3.

$$\begin{aligned}
 & 5^{k+1} + 2 \times 11^{k+1} \\
 &= 5 \times 5^k + 2 \times 11^{k+1} \\
 &= 5 \times (3m - 2 \times 11^k) + 2 \times 11^{k+1} && \text{by the induction hypothesis (**)} \\
 &= 15m - 10 \times 11^k + 2 \times 11 \times 11^k \\
 &= 15m - 10 \times 11^k + 22 \times 11^k \\
 &= 15m + 12 \times 11^k \\
 &= 3(5m + 4 \times 11^k) \text{ which is divisible by 3.}
 \end{aligned}$$

Chapter 2 worked solutions – Mathematical induction

C

It follows from parts **A** and **B** by mathematical induction that the statement is true for all integers $n \geq 0$.

5b **A**

When $n = 0$, $3^{3(0)} + 2^{0+2} = 1 + 2^2 = 1 + 4 = 5$ which is divisible by 5

so the statement is true for $n = 0$.

B

Suppose that $k \geq 0$ is an integer for which the statement is true.

That is, suppose $3^{3k} + 2^{k+2} = 5m$, for some integer m .

Note that rearranging this gives $3^{3k} = 5m - 2^{k+2}$. (**)

We prove the statement for $n = k + 1$.

That is, we prove $3^{3(k+1)} + 2^{(k+1)+2}$ is divisible by 5.

$$\begin{aligned}
 & 3^{3(k+1)} + 2^{(k+1)+2} \\
 &= 3^{3k+3} + 2^{k+3} \\
 &= 3^3 \times 3^{3k} + 2 \times 2^{k+2} \\
 &= 27 \times 3^{3k} + 2 \times 2^{k+2} \\
 &= 27(5m - 2^{k+2}) + 2 \times 2^{k+2} \quad \text{by the induction hypothesis (**)} \\
 &= 27 \times 5m - 27 \times 2^{k+2} + 2 \times 2^{k+2} \\
 &= 27 \times 5m - 25 \times 2^{k+2} \\
 &= 5(27 \times m - 5 \times 2^{k+2}) \text{ which is divisible by 5.}
 \end{aligned}$$

C

It follows from parts **A** and **B** by mathematical induction that the statement is true for all integers $n \geq 0$.

5c **A**

When $n = 0$, $11^{0+2} + 12^{0+1} = 11^2 + 12 = 133$ which is divisible by 133.

So, the statement is true for $n = 0$.

B

Suppose that $k \geq 0$ is an integer for which the statement is true

That is, suppose $11^{k+2} + 12^{2k+1} = 133m$, for some integer m .

Note that rearranging this gives $11^{k+2} = 133m - 12^{2k+1}$. (**)

Chapter 2 worked solutions – Mathematical induction

We prove the statement for $n = k + 1$.

That is, we prove $11^{(k+1)+2} + 12^{2(k+1)+1}$ is divisible by 133.

$$\begin{aligned}
 & 11^{(k+1)+2} + 12^{2(k+1)+1} \\
 &= 11^{k+3} + 12^{2k+3} \\
 &= 11 \times 11^{k+2} + 12^2 \times 12^{2k+1} \\
 &= 11 \times (133m - 12^{2k+1}) + 12^2 \times 12^{2k+1} \quad \text{by the induction hypothesis (**)} \\
 &= 11 \times 133m - 11 \times 12^{2k+1} + 12^2 \times 12^{2k+1} \\
 &= 11 \times 133m + (12^2 - 11)12^{2k+1} \\
 &= 11 \times 133m + 133 \times 12^{2k+1} \\
 &= 133(11m + 12^{2k+1}) \text{ which is divisible by 133.}
 \end{aligned}$$

C

It follows from parts **A** and **B** by mathematical induction that the statement is true for all integers $n \geq 0$.

6 **A**

When $n = 1$, $x^1 - 1 = x - 1$ which is divisible by $x - 1$.

So, the statement is true for $n = 1$.

B

Suppose that $k \geq 1$ is a positive integer for which the statement is true.

That is, suppose $x^k - 1 = m(x - 1)$, for some integer m . (**)

We prove the statement for $n = k + 1$.

That is, we prove $x^{k+1} - 1$ is divisible by $x - 1$.

$$\begin{aligned}
 & x^{k+1} - 1 \\
 &= x \times x^k - 1 \\
 &= x(x^k - 1) + x - 1 \\
 &= xm(x - 1) + (x - 1) \quad \text{by the induction hypothesis (**)} \\
 &= (x - 1)(xm + 1), \text{ which is divisible by } x - 1
 \end{aligned}$$

C

It follows from parts **A** and **B** by mathematical induction that the statement is true for all integers $n \geq 1$.

Chapter 2 worked solutions – Mathematical induction

7 Suppose that $k \geq 1$ is a positive integer for which the statement is true.

That is, suppose $8k^2 + 14 = 4m$, for some integer m . (**)

We prove the statement for $n = k + 1$.

That is, we prove $8(k + 1)^2 + 14$ is divisible by 4.

$$\begin{aligned}
 &8(k + 1)^2 + 14 \\
 &= 8(k^2 + 2k + 1) + 14 \\
 &= 8k^2 + 16k + 8 + 14 \\
 &= (8k^2 + 14) + 16k + 8 \\
 &= 4m + 16k + 8 && \text{by the induction hypothesis (**)} \\
 &= 4(m + 4k + 2) \text{ which is divisible by 4.}
 \end{aligned}$$

Now we show that $8n^2 + 14$ is never divisible by 4 if n is an integer.

Firstly, we show by induction that $8n^2 + 14 = 4a + 2$ for some integer a .

When $n = 1$, $8(1) + 14 = 22 = 4 \times 5 + 2$ which is in the form $4a + 2$.

So, the statement is true for $n = 1$.

Suppose that $k \geq 1$ is a positive integer for which the statement is true.

That is, suppose $8k^2 + 14 = 4m + 2$, for some integer m . (**)

We prove the statement for $n = k + 1$.

That is, we prove $8(k + 1)^2 + 14 = 4a + 2$ for some integer a .

$$\begin{aligned}
 &8(k + 1)^2 + 14 \\
 &= 8(k^2 + 2k + 1) + 14 \\
 &= 8k^2 + 16k + 8 + 14 \\
 &= (8k^2 + 14) + 16k + 8 \\
 &= 4m + 2 + 16k + 8 && \text{by the induction hypothesis (**)} \\
 &= 4(m + 4k + 2) + 2 \text{ which is in the form } 4a + 2 \text{ where } a \text{ is an integer}
 \end{aligned}$$

$4a + 2$ always has remainder 2 upon division by 4 and hence is never divisible by 4 (assuming that n was a whole number).

This shows that the first step of the proof, showing true for a base case such as $n = 0$ or $n = 1$, is necessary.

Chapter 2 worked solutions – Mathematical induction

8a Show that the statement is true for $n = 0$.

$$f(0) = 0 - 0 + 17 = 17 \text{ which is prime.}$$

Suppose that $k \geq 0$ is an integer for which the statement is true.

That is, suppose $k^2 - k + 17 = P$, where P is a prime. (**)

We attempt to prove the statement for $n = k + 1$.

That is, we attempt to prove $(k + 1)^2 - (k + 1) + 17$ is prime.

$$\begin{aligned} & (k + 1)^2 - (k + 1) + 17 \\ &= k^2 + 2k + 1 - k - 1 + 17 \\ &= (k^2 - k + 17) + 2k \\ &= P + 2k \end{aligned} \quad \text{by the induction hypothesis (**)}$$

And now we are stuck!

Hence, we require the step showing that the statement is true for $n = k + 1$ given that it is true for any $n = k$. Below we list the first 17 outputs of the function. Note that the first 16 are prime.

n	0	1	2	3	4	5	6	7	8	9
$f(n)$	17	17	19	23	29	37	47	59	73	89

n	10	11	12	13	14	15	16	17
$f(n)$	107	127	149	173	199	227	257	289

Note that $289 = 17^2$ and hence it is not prime.

8b Show that the statement is true for $n = 0$.

$$f(0) = 0 + 0 + 41 = 41 \text{ which is prime.}$$

Suppose that $k \geq 0$ is an integer for which the statement is true.

That is, suppose $k^2 + k + 41 = P$, where P is a prime. (**)

We attempt to prove the statement for $n = k + 1$.

That is, we attempt to prove $(k + 1)^2 + (k + 1) + 41$ is prime.

$$\begin{aligned} & (k + 1)^2 + (k + 1) + 41 \\ &= k^2 + 2k + 1 + k + 1 + 41 \\ &= (k^2 + k + 41) + 2k + 2 \end{aligned}$$

Chapter 2 worked solutions – Mathematical induction

$$= P + 2k + 2$$

by the induction hypothesis (**)

And now we are stuck!

Hence, we require the step showing that the statement is true for $n = k + 1$ given that it is true for any $n = k$.

Note that $f(40) = 1681 = 41^2$ and hence is not prime, thus providing us with a counter-example.

9 **A**

When $n = 0$, $3^{2^0} - 1 = 3^1 - 1 = 3 - 1 = 2$ which is divisible by 2^1
so the statement is true for $n = 0$.

B

Suppose that $k \geq 0$ is an integer for which the statement is true.

That is, suppose $3^{2^k} - 1 = 2^{k+1}m$, for some positive integer m .

Note that rearranging this gives $3^{2^k} = 2^{k+1}m + 1$. (**)

We prove the statement for $n = k + 1$.

That is, we prove $3^{2^{k+1}} - 1$ is divisible by 2^{k+2} .

$$\begin{aligned} & 3^{2^{k+1}} - 1 \\ &= 3^{2 \times 2^k} - 1 \\ &= (3^{2^k})^2 - 1 \\ &= (2^{k+1}m + 1)^2 - 1 && \text{by the induction hypothesis (**)} \\ &= 2^{2k+2}m^2 + 2^{k+2}m + 1 - 1 \\ &= 2^{k+2}(2^k m^2 + m), \text{ which is divisible by } 2^{k+2}. \end{aligned}$$

C

It follows from parts **A** and **B** by mathematical induction that the statement is true for all integers $n \geq 0$.

10 **A**

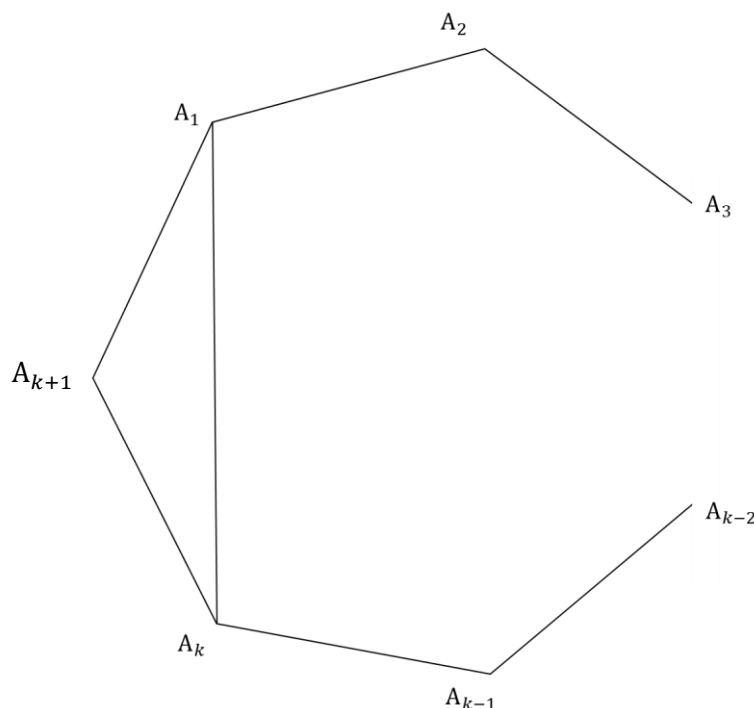
When $n = 3$, $(3 - 2) \times 180^\circ = 180^\circ$ which is the angle sum of a triangle or 1 straight angle so the statement is true for $n = 3$.

Chapter 2 worked solutions – Mathematical induction

B

Assume that a k -gon, where $k \geq 3$, has angle sum $(k - 2) \times 180^\circ$. (**)

Prove that a $(k + 1)$ -gon has angle sum $(k - 1) \times 180^\circ$.



The angle sum of the $(k + 1)$ -gon $A_1A_2 \dots A_kA_{k+1}$
 = the angle sum of the k -gon $A_1A_2 \dots A_k$ + the angle sum of $\triangle A_1A_kA_{k+1}$
 = $(k - 2) \times 180^\circ + 180^\circ$ by the induction hypothesis (**)
 = $(k - 1) \times 180^\circ$ as required.

C

It follows from parts **A** and **B** by mathematical induction that the statement is true for all integers $n \geq 3$.

11 **A**

A 0-member set is the empty set which has $2^0 = 1$ subset (itself) so the result is true for $n = 0$.

B

Assume that a k -member set has 2^k subsets. (**)

Prove that a $(k + 1)$ -member set has 2^{k+1} subsets.

Chapter 2 worked solutions – Mathematical induction

Suppose we have a k -member set, and we add a new member.

Then each subset of the k -member set (there are 2^k of these by (**)) is also a subset of the $(k + 1)$ -member set, and we get 2^k new subsets when we add the new member to each of the previous 2^k subsets.

So the resulting number of subsets for the $(k + 1)$ -member set is

$$2^k + 2^k = 2 \times 2^k = 2^{k+1} \text{ as required.}$$

C

It follows from parts **A** and **B** by mathematical induction that the statement is true for all integers $n \geq 0$.

12 **A**

$$\frac{d}{dx}(x^1) = 1 = 1x^0 \text{ so the result is true for } n = 1.$$

B

$$\text{Assume that } \frac{d}{dx}(x^k) = kx^{k-1}. \quad (**)$$

$$\text{Prove that } \frac{d}{dx}(x^{k+1}) = (k + 1)x^k.$$

$$\begin{aligned} \text{LHS} &= \frac{d}{dx}(x \times x^k) \\ &= x \times \frac{d}{dx}(x^k) + x^k \times \frac{d}{dx}(x) && \text{by the product rule} \\ &= x \times kx^{k-1} + x^k \times 1 && \text{by the induction hypothesis (**)} \\ &= kx^k + x^k \\ &= (k + 1)x^k \\ &= \text{RHS} \end{aligned}$$

C

It follows from parts **A** and **B** by mathematical induction that the statement is true for all integers $n \geq 1$.

Chapter 2 worked solutions – Mathematical induction

13a **A**

When $n = 0$, $n^2 + 2n = 0^2 + 2(0) = 0$ which is a multiple of 8

so the result is true for $n = 0$.

B

Assume that the result is true for $n = k$, where k is even.

That is, assume that $k^2 + 2k = 8m$, where m is a positive integer. (**)

Prove the result is true for $n = k + 2$.

That is, prove that $(k + 2)^2 + 2(k + 2)$ is a multiple of 8.

$$\begin{aligned}
 & (k + 2)^2 + 2(k + 2) \\
 &= k^2 + 4k + 4 + 2k + 4 \\
 &= (k^2 + 2k) + 4k + 8 \\
 &= 8m + 4k + 8 && \text{by the induction hypothesis (**)} \\
 &= 8m + 4 \times 2l + 8 && (k \text{ is even, so } k = 2l \text{ for some integer } l) \\
 &= 8(m + l + 1) \text{ which is a multiple of 8.}
 \end{aligned}$$

C

It follows from parts **A** and **B** by mathematical induction that the statement is true for all even integers $n \geq 0$.

13b **A**

When $n = 1$, $3^1 + 7^1 = 10$ which is divisible by 10

so the result is true for $n = 1$.

B

Assume that the result is true for $n = k$, where k is odd.

That is, assume that $3^k + 7^k = 10m$, where m is a positive integer.

This can be rearranged as $3^k = 10m - 7^k$. (**)

Prove the result is true for $n = k + 2$.

That is, prove that $3^{k+2} + 7^{k+2}$ is a multiple of 10.

$$\begin{aligned}
 & 3^{k+2} + 7^{k+2} \\
 &= 3^2 \times 3^k + 7^2 \times 7^k \\
 &= 9 \times 3^k + 49 \times 7^k \\
 &= 9 \times (10m - 7^k) + 49 \times 7^k && \text{by the induction hypothesis (**)} \\
 &= 90m - 9 \times 7^k + 49 \times 7^k
 \end{aligned}$$

Chapter 2 worked solutions – Mathematical induction

$$= 90m + 40 \times 7^k$$

$$= 10(9m + 4 \times 7^k) \text{ which is divisible by 10.}$$

C

It follows from parts **A** and **B** by mathematical induction that the statement is true for all odd integers $n \geq 1$.

Chapter 2 worked solutions – Mathematical induction

Solutions to Chapter 2 Review

1a **A**When $n = 1$,

$$\begin{aligned}\text{RHS} &= 1(2(1) - 1) \\ &= 1 \\ &= \text{LHS}\end{aligned}$$

so the statement is true for $n = 1$.**B**Suppose that $k \geq 1$ is a positive integer for which the statement is true.That is, suppose $1 + 5 + 9 + \cdots + (4k - 3) = k(2k - 1)$ (**)We prove the statement for $n = k + 1$.

That is, we prove

$$\begin{aligned}1 + 5 + 9 + \cdots + (4k - 3) + (4(k + 1) - 3) &= (k + 1)(2(k + 1) - 1) \\ \text{LHS} &= 1 + 5 + 9 + \cdots + (4k - 3) + (4(k + 1) - 3) \\ &= 1 + 5 + 9 + \cdots + (4k - 3) + (4k + 4 - 3) \\ &= 1 + 5 + 9 + \cdots + (4k - 3) + (4k + 1) \\ &= k(2k - 1) + (4k + 1) && \text{by the induction hypothesis (**)} \\ &= 2k^2 - k + 4k + 1 \\ &= 2k^2 + 3k + 1 \\ &= (k + 1)(2k + 1) \\ &= (k + 1)(2(k + 1) - 1) \\ &= \text{RHS}\end{aligned}$$

CIt follows from parts **A** and **B** by mathematical induction, that:

$$1 + 5 + 9 + \cdots + (4n - 3) = n(2n - 1) \text{ for all integers } n \geq 1.$$

1b **A**When $n = 1$,

$$\begin{aligned}\text{RHS} &= \frac{1}{6}(7^1 - 1) \\ &= \frac{6}{6} \\ &= 1 \\ &= \text{LHS}\end{aligned}$$

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so the statement is true for $n = 1$.

B

Suppose that $k \geq 1$ is a positive integer for which the statement is true.

That is, suppose $1 + 7 + 7^2 + \dots + 7^{k-1} = \frac{1}{6}(7^k - 1)$ (**)

We prove the statement for $n = k + 1$.

That is, we prove $1 + 7 + 7^2 + \dots + 7^{k-1} + 7^{k+1-1} = \frac{1}{6}(7^{k+1} - 1)$

$$\begin{aligned}
 \text{LHS} &= 1 + 7 + 7^2 + \dots + 7^{k-1} + 7^{k+1-1} \\
 &= 1 + 7 + 7^2 + \dots + 7^{k-1} + 7^k \\
 &= \frac{1}{6}(7^k - 1) + 7^k && \text{by the induction hypothesis (**)} \\
 &= \frac{1}{6}(7^k - 1 + 6 \times 7^k) \\
 &= \frac{1}{6}(7 \times 7^k - 1) \\
 &= \frac{1}{6}(7^{k+1} - 1) \\
 &= \text{RHS}
 \end{aligned}$$

C

It follows from parts **A** and **B** by mathematical induction, that:

$$1 + 7 + 7^2 + \dots + 7^{n-1} = \frac{1}{6}(7^n - 1) \text{ for all integers } n \geq 1.$$

1c **A**

When $n = 1$,

$$\begin{aligned}
 \text{RHS} &= \frac{1}{6}(1)(1+1)(2 \times 1 + 13) \\
 &= \frac{1}{6}(1)(2)(15) \\
 &= 5 \\
 &= \text{LHS}
 \end{aligned}$$

so the statement is true for $n = 1$.

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B

Suppose that $k \geq 1$ is a positive integer for which the statement is true.

That is, suppose $1 \times 5 + 2 \times 6 + 3 \times 7 + \cdots + k(k+4)$

$$= \frac{1}{6}k(k+1)(2k+13) \quad (**)$$

We prove the statement for $n = k + 1$.

That is, we prove $1 \times 5 + 2 \times 6 + 3 \times 7 + \cdots + k(k+4) + (k+1)((k+1)+4)$

$$= \frac{1}{6}(k+1)((k+1)+1)(2(k+1)+13)$$

$$\text{LHS} = 1 \times 5 + 2 \times 6 + 3 \times 7 + \cdots + k(k+4) + (k+1)((k+1)+4)$$

$$= \frac{1}{6}k(k+1)(2k+13) + (k+1)((k+1)+4)$$

by the induction hypothesis (**)

$$= \frac{1}{6}(k+1)[k(2k+13) + 6((k+1)+4)]$$

$$= \frac{1}{6}(k+1)[k(2k+13) + 6(k+5)]$$

$$= \frac{1}{6}(k+1)[2k^2 + 13k + 6k + 30]$$

$$= \frac{1}{6}(k+1)[2k^2 + 19k + 30]$$

$$= \frac{1}{6}(k+1)(k+2)(2k+15)$$

$$= \frac{1}{6}(k+1)((k+1)+1)(2(k+1)+13)$$

$$= \text{RHS}$$

C

It follows from parts **A** and **B** by mathematical induction, that:

$$1 \times 5 + 2 \times 6 + 3 \times 7 + \cdots + n(n+4) = \frac{1}{6}n(n+1)(2n+13) \text{ for all integers } n \geq 1.$$

1d **A**

When $n = 1$,

$$\text{RHS} = \frac{1}{2(1+2)}$$

$$= \frac{1}{2(3)}$$

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$$\begin{aligned}
 &= \frac{1}{6} \\
 &= \text{LHS}
 \end{aligned}$$

so the statement is true for $n = 1$.

B

Suppose that $k \geq 1$ is a positive integer for which the statement is true.

$$\text{That is, suppose } \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \frac{1}{4 \times 5} + \cdots + \frac{1}{(k+1)(k+2)} = \frac{k}{2(k+2)} \quad (**)$$

We prove the statement for $n = k + 1$.

$$\text{That is, we prove } \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \frac{1}{4 \times 5} + \cdots + \frac{1}{(k+1)(k+2)} + \frac{1}{((k+1)+1)((k+1)+2)} = \frac{(k+1)}{2((k+1)+2)}$$

$$\begin{aligned}
 \text{LHS} &= \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \frac{1}{4 \times 5} + \cdots + \frac{1}{(k+1)(k+2)} + \frac{1}{((k+1)+1)((k+1)+2)} \\
 &= \frac{k}{2(k+2)} + \frac{1}{((k+1)+1)((k+1)+2)} \quad \text{by the induction hypothesis (**)} \\
 &= \frac{k}{2(k+2)} + \frac{2}{2((k+1)+1)((k+1)+2)} \\
 &= \frac{k}{2(k+2)} + \frac{2}{2(k+2)(k+3)} \\
 &= \frac{k(k+3)}{2(k+2)(k+3)} + \frac{2}{2(k+2)(k+3)} \\
 &= \frac{k^2 + 3k + 2}{2(k+2)(k+3)} \\
 &= \frac{(k+1)(k+2)}{2(k+2)(k+3)} \\
 &= \frac{(k+1)}{2(k+3)} \\
 &= \frac{(k+1)}{2((k+1)+2)} \\
 &= \text{RHS}
 \end{aligned}$$

C

It follows from parts **A** and **B** by mathematical induction, that:

$$\frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \frac{1}{4 \times 5} + \cdots + \frac{1}{(n+1)(n+2)} = \frac{n}{2(n+2)} \text{ for all integers } n \geq 1.$$

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1e **A**When $n = 1$,

$$\begin{aligned}
 \text{RHS} &= 2 - \frac{1+2}{2^1} \\
 &= 2 - \frac{3}{2} \\
 &= \frac{1}{2} \\
 &= \text{LHS}
 \end{aligned}$$

so the statement is true for $n = 1$.**B**Suppose that $k \geq 1$ is a positive integer for which the statement is true.

$$\text{That is, suppose } \frac{1}{2} + \frac{2}{2^2} + \frac{3}{2^3} + \cdots + \frac{k}{2^k} = 2 - \frac{k+2}{2^k} \quad (**)$$

We prove the statement for $n = k + 1$.

$$\text{That is, we prove } \frac{1}{2} + \frac{2}{2^2} + \frac{3}{2^3} + \cdots + \frac{k}{2^k} + \frac{k+1}{2^{k+1}} = 2 - \frac{k+3}{2^{k+1}}$$

$$\begin{aligned}
 \text{LHS} &= \frac{1}{2} + \frac{2}{2^2} + \frac{3}{2^3} + \cdots + \frac{k}{2^k} + \frac{k+1}{2^{k+1}} \\
 &= 2 - \frac{k+2}{2^k} + \frac{k+1}{2^{k+1}} && \text{by the induction hypothesis (**)} \\
 &= 2 - \frac{2(k+2) - (k+1)}{2^{k+1}} \\
 &= 2 - \frac{2k+4-k-1}{2^{k+1}} \\
 &= 2 - \frac{k+3}{2^{k+1}} \\
 &= \text{RHS}
 \end{aligned}$$

CIt follows from parts **A** and **B** by mathematical induction, that:

$$\frac{1}{2} + \frac{2}{2^2} + \frac{3}{2^3} + \cdots + \frac{n}{2^n} = 2 - \frac{n+2}{2^n} \text{ for all integers } n \geq 1.$$

2a **A**When $n = 1$, $7^{2-1} + 5 = 7 + 5 = 12$ which is divisible by 12so the statement is true for $n = 1$.

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B

Suppose that $k \geq 1$ is a positive integer for which the statement is true.

That is, suppose $7^{2k-1} + 5 = 12m$, for some integer m .

Note that rearranging this gives $7^{2k-1} = 12m - 5$. (**)

We prove the statement for $n = k + 1$.

That is, we prove $7^{2(k+1)-1} + 5$ is divisible by 12.

$$\begin{aligned}
 & 7^{2(k+1)-1} + 5 \\
 &= 7^{2k+2-1} + 5 \\
 &= 7^{2k+1} + 5 \\
 &= 7^2 \times 7^{2k-1} + 5 \\
 &= 7^2(12m - 5) + 5 && \text{by the induction hypothesis (**)} \\
 &= 49(12m - 5) + 5 \\
 &= 49 \times 12m - 5 \times 49 + 5 \times 1 \\
 &= 49 \times 12m + 5(1 - 49) \\
 &= 49 \times 12m - 5 \times 12 \times 4 \\
 &= 12(49m - 20) \text{ which is divisible by 12.}
 \end{aligned}$$

C

It follows from parts **A** and **B** by mathematical induction that the statement is true for all integers $n \geq 1$.

2b **A**

When $n = 0$, $2^0 + 6(0) - 1 = 1 + 0 - 1 = 0$ which is divisible by 9

so the statement is true for $n = 0$.

B

Suppose that $k \geq 0$ is an integer for which the statement is true.

That is, suppose $2^{2k} + 6k - 1 = 9m$, for some integer m .

Note that rearranging this gives $2^{2k} = 9m - 6k + 1$. (**)

We prove the statement for $n = k + 1$.

That is, we prove $2^{2(k+1)} + 6(k+1) - 1$ is divisible by 9.

$$\begin{aligned}
 & 2^{2(k+1)} + 6(k+1) - 1 \\
 &= 2^{2k+2} + 6k + 6 - 1 \\
 &= 2^2 \times 2^{2k} + 6k + 5
 \end{aligned}$$

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$$\begin{aligned}
 &= 2^2 \times (9m - 6k + 1) + 6k + 5 \quad \text{by the induction hypothesis (**)} \\
 &= 36m - 24k + 4 + 6k + 5 \\
 &= 36m - 18k + 9 \\
 &= 9(4m - 2k + 1) \text{ which is divisible by 9.}
 \end{aligned}$$

C

It follows from parts **A** and **B** by mathematical induction that the statement is true for all integers $n \geq 0$.

2c **A**

When $n = 1$, $2^4 + 5^1 = 16 + 5 = 21$ which is divisible by 21.

so the statement is true for $n = 1$.

B

Suppose that $k \geq 1$ is a positive integer for which the statement is true.

That is, suppose $2^{2k+2} + 5^{2k-1} = 21m$, for some integer m .

Note that rearranging this gives $2^{2k+2} = 21m - 5^{2k-1}$. (**)

We prove the statement for $n = k + 1$.

That is, we prove $2^{2(k+1)+2} + 5^{2(k+1)-1}$ is divisible by 21.

$$\begin{aligned}
 &2^{2(k+1)+2} + 5^{2(k+1)-1} \\
 &= 2^{2k+2+2} + 5^{2k-1+2} \\
 &= 2^2 \times 2^{2k+2} + 5^2 \times 5^{2k-1} \\
 &= 2^2 \times (21m - 5^{2k-1}) + 5^2 \times 5^{2k-1} \quad \text{by the induction hypothesis (**)} \\
 &= 4 \times (21m - 5^{2k-1}) + 5^2 \times 5^{2k-1} \\
 &= 4 \times 21m - 4 \times 5^{2k-1} + 5^2 \times 5^{2k-1} \\
 &= 4 \times 21m - 4 \times 5^{2k-1} + 25 \times 5^{2k-1} \\
 &= 4 \times 21m + 21 \times 5^{2k-1} \\
 &= 21(4m + 5^{2k-1}) \text{ which is divisible by 21.}
 \end{aligned}$$

C

It follows from parts **A** and **B** by mathematical induction that the statement is true for all integers $n \geq 1$.

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2d **A**

When $n = 0$, $0 + 1^3 + 2^3 = 0 + 1 + 8 = 9$ which is divisible by 9

so the statement is true for $n = 0$.

B

Suppose that $k \geq 0$ is an integer for which the statement is true.

That is, suppose $k^3 + (k + 1)^3 + (k + 2)^3 = 9m$, for some integer m .

Note that rearranging this gives $(k + 1)^3 + (k + 2)^3 = 9m - k^3$ (**)

We prove the statement for $n = k + 1$.

That is, we prove $(k + 1)^3 + ((k + 1) + 1)^3 + ((k + 1) + 2)^3$ is divisible by 9.

$$\begin{aligned}
 & (k + 1)^3 + ((k + 1) + 1)^3 + ((k + 1) + 2)^3 \\
 &= (k + 1)^3 + (k + 2)^3 + (k + 3)^3 \\
 &= 9m - k^3 + (k + 3)^3 && \text{by the induction hypothesis (**)} \\
 &= 9m - k^3 + k^3 + 9k^2 + 27k + 27 \\
 &= 9m + 9k^2 + 27k + 27 \\
 &= 9(m + k^2 + 3k + 3) \text{ which is divisible by 9.}
 \end{aligned}$$

C

It follows from parts **A** and **B** by mathematical induction that the statement is true for all integers $n \geq 0$.

3a

n	0	1	2	3
$2^{3n} - 3^n$	0	5	55	485

The expression is always divisible by 5 for all whole numbers $n \geq 0$.

3b **A**

When $n = 0$, $2^0 - 3^0 = 1 - 1 = 0$ which is divisible by 5

so the statement is true for $n = 0$.

B

Suppose that $k \geq 0$ is an integer for which the statement is true.

That is, suppose $2^{3k} - 3^k = 5m$, for some integer m .

Note that rearranging this gives $2^{3k} = 5m + 3^k$. (**)

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We prove the statement for $n = k + 1$.

That is, we prove $2^{3(k+1)} - 3^{k+1}$ is divisible by 5.

$$\begin{aligned}
 & 2^{3(k+1)} - 3^{k+1} \\
 &= 2^{3k+3} - 3^{k+1} \\
 &= 2^3 \times 2^{3k} - 3 \times 3^k \\
 &= 2^3 \times (5m + 3^k) - 3 \times 3^k && \text{by the induction hypothesis (**)} \\
 &= 8 \times (5m + 3^k) - 3 \times 3^k \\
 &= 5 \times 8m + 8 \times 3^k - 3 \times 3^k \\
 &= 5 \times 8m + 5 \times 3^k \\
 &= 5(8m + 3^k) \text{ which is divisible by 5.}
 \end{aligned}$$

C

It follows from parts **A** and **B** by mathematical induction that the statement is true for all whole numbers $n \geq 0$.

4a **A**

When $n = 1$,

$$\begin{aligned}
 \text{RHS} &= (1 + 1)! - 1 \\
 &= 2! - 1 \\
 &= 2 - 1 \\
 &= 1
 \end{aligned}$$

$$\begin{aligned}
 \text{LHS} &= \sum_{r=1}^1 r \times r! \\
 &= 1 \times 1! \\
 &= 1 \\
 &= \text{RHS}
 \end{aligned}$$

so the statement is true for $n = 1$.

B

Suppose that $k \geq 1$ is a positive integer for which the statement is true.

$$\text{That is, suppose } \sum_{r=1}^k r \times r! = (k + 1)! - 1 \quad (**)$$

We prove the statement for $n = k + 1$.

$$\text{That is, we prove } \sum_{r=1}^{k+1} r \times r! = ((k + 1) + 1)! - 1.$$

$$\text{LHS} = \sum_{r=1}^{k+1} r \times r!$$

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$$\begin{aligned}
&= \sum_{r=1}^k r \times r! + (k+1) \times (k+1)! \\
&= (k+1)! - 1 + (k+1) \times (k+1)! \quad \text{by the induction hypothesis (**),} \\
&= (k+1)! + (k+1) \times (k+1)! - 1 \\
&= (1 + (k+1))(k+1)! - 1 \\
&= (k+2)(k+1)! - 1 \\
&= (k+2)! - 1 \\
&= ((k+1) + 1)! - 1 \\
&= \text{RHS}
\end{aligned}$$

C

It follows from parts **A** and **B** by mathematical induction, that:

$$\sum_{r=1}^n r \times r! = (n+1)! - 1 \quad \text{for all integers } n \geq 1.$$

4b **A**

When $n = 1$,

$$\begin{aligned}
\text{RHS} &= 1 - \frac{1}{1!} \\
&= 1 - 1 \\
&= 0 \\
\text{LHS} &= \sum_{r=1}^1 \frac{r-1}{r!} \\
&= \frac{1-1}{1!} \\
&= 0 \\
&= \text{RHS}
\end{aligned}$$

so the statement is true for $n = 1$.

B

Suppose that $k \geq 1$ is a positive integer for which the statement is true.

$$\text{That is, suppose } \sum_{r=1}^k \frac{r-1}{r!} = 1 - \frac{1}{k!} \quad (**)$$

We prove the statement for $n = k+1$.

$$\text{That is, we prove } \sum_{r=1}^{k+1} \frac{r-1}{r!} = 1 - \frac{1}{(k+1)!}.$$

$$\text{LHS} = \sum_{r=1}^{k+1} \frac{r-1}{r!}$$

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$$\begin{aligned}
 &= \sum_{r=1}^k \frac{r-1}{r!} + \frac{(k+1)-1}{(k+1)!} \\
 &= 1 - \frac{1}{k!} + \frac{(k+1)-1}{(k+1)!} \quad \text{by the induction hypothesis (**),} \\
 &= 1 - \frac{k+1}{(k+1)!} + \frac{(k+1)-1}{(k+1)!} \\
 &= 1 - \frac{k+1-k}{(k+1)!} \\
 &= 1 - \frac{1}{(k+1)!} \\
 &= \text{RHS}
 \end{aligned}$$

C

It follows from parts **A** and **B** by mathematical induction, that:

$$\sum_{r=1}^n \frac{r-1}{r!} = 1 - \frac{1}{n!} \text{ for all integers } n \geq 1.$$

The limiting sum of the series in part b is:

$$\begin{aligned}
 &\lim_{n \rightarrow \infty} \left(1 - \frac{1}{n!} \right) \\
 &= 1
 \end{aligned}$$

5 A

When $n = 1$,

$$\begin{aligned}
 \text{RHS} &= \frac{1}{2}(1)(6 - 3 - 1) \\
 &= \frac{2}{2} \\
 &= 1
 \end{aligned}$$

$$\text{LHS} = 1^2$$

$$= 1$$

$$= \text{RHS}$$

so the statement is true for $n = 1$.

B

Suppose that $k \geq 1$ is a positive integer for which the statement is true.

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That is, suppose $1^2 + 4^2 + 7^2 + \dots + (3k - 2)^2 = \frac{1}{2}k(6k^2 - 3k - 1)$ (**)

We prove the statement for $n = k + 1$.

That is, we prove $1^2 + 4^2 + 7^2 + \dots + (3k - 2)^2 + (3(k + 1) - 2)^2$
 $= \frac{1}{2}(k + 1)(6(k + 1)^2 - 3(k + 1) - 1)$

$$\begin{aligned}
 \text{LHS} &= 1^2 + 4^2 + 7^2 + \dots + (3k - 2)^2 + (3(k + 1) - 2)^2 \\
 &= \frac{1}{2}k(6k^2 - 3k - 1) + (3(k + 1) - 2)^2 \quad \text{by the induction hypothesis (**)} \\
 &= \frac{1}{2}k(6k^2 - 3k - 1) + (3k + 3 - 2)^2 \\
 &= \frac{1}{2}k(6k^2 - 3k - 1) + (3k + 1)^2 \\
 &= \frac{1}{2}(6k^3 - 3k^2 - k) + 9k^2 + 6k + 1 \\
 &= \frac{1}{2}(6k^3 - 3k^2 - k + 18k^2 + 12k + 2) \\
 &= \frac{1}{2}(6k^3 + 15k^2 + 11k + 2) \\
 &= \frac{1}{2}(k + 1)(6k^2 + 9k + 2) \\
 \text{RHS} &= \frac{1}{2}(k + 1)(6(k + 1)^2 - 3(k + 1) - 1) \\
 &= \frac{1}{2}(k + 1)(6(k^2 + 2k + 1) - 3k - 3 - 1) \\
 &= \frac{1}{2}(k + 1)(6k^2 + 12k + 6 - 3k - 4) \\
 &= \frac{1}{2}(k + 1)(6k^2 + 9k + 2) \\
 &= \text{LHS}
 \end{aligned}$$

C

It follows from parts **A** and **B** by mathematical induction, that:

$$1^2 + 4^2 + 7^2 + \dots + (3n - 2)^2 = \frac{1}{2}n(6n^2 - 3n - 1) \text{ for all integers } n \geq 1.$$

6 A

When $n = 1$,

$$\begin{aligned}
 \text{RHS} &= 1^2(2 - 1) \\
 &= 1 \\
 &= 1^3 \\
 &= \text{LHS}
 \end{aligned}$$

so the statement is true for $n = 1$.

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B

Suppose that $k \geq 1$ is a positive integer for which the statement is true.

That is, suppose $1^3 + 3^3 + 5^3 + \dots + (2k - 1)^3 = k^2(2k^2 - 1)$ (**)

We prove the statement for $n = k + 1$.

That is, we prove $1^3 + 3^3 + 5^3 + \dots + (2k - 1)^3 + (2(k + 1) - 1)^3$
 $= (k + 1)^2(2(k + 1)^2 - 1)$

$$\begin{aligned}
 \text{LHS} &= 1^3 + 3^3 + 5^3 + \dots + (2k - 1)^3 + (2(k + 1) - 1)^3 \\
 &= k^2(2k^2 - 1) + (2(k + 1) - 1)^3 \quad \text{by the induction hypothesis (**)} \\
 &= k^2(2k^2 - 1) + (2k + 2 - 1)^3 \\
 &= k^2(2k^2 - 1) + (2k + 1)^3 \\
 &= k^2(2k^2 - 1) + 8k^3 + 12k^2 + 6k + 1 \\
 &= 2k^4 - k^2 + 8k^3 + 12k^2 + 6k + 1 \\
 &= 2k^4 + 8k^3 + 11k^2 + 6k + 1 \\
 &= (k + 1)^2(2k^2 + 4k + 1) \\
 &= (k + 1)^2(2(k^2 + 2k + 1) - 1) \\
 &= (k + 1)^2(2(k + 1)^2 - 1) \\
 &= \text{RHS}
 \end{aligned}$$

C

It follows from parts **A** and **B** by mathematical induction, that:

$$1^3 + 3^3 + 5^3 + \dots + (2n - 1)^3 = n^2(2n^2 - 1) \quad \text{for all integers } n \geq 1.$$